

REST APIs - REST Web Services



You will learn how to ...

- Create REST APIs / Web Services with Spring
- Discuss REST concepts, JSON and HTTP messaging
- Install REST client tool: Postman
- Develop REST APIs / Web Services with **@RestController**
- Build a CRUD interface to the database with Spring REST

Practical Results

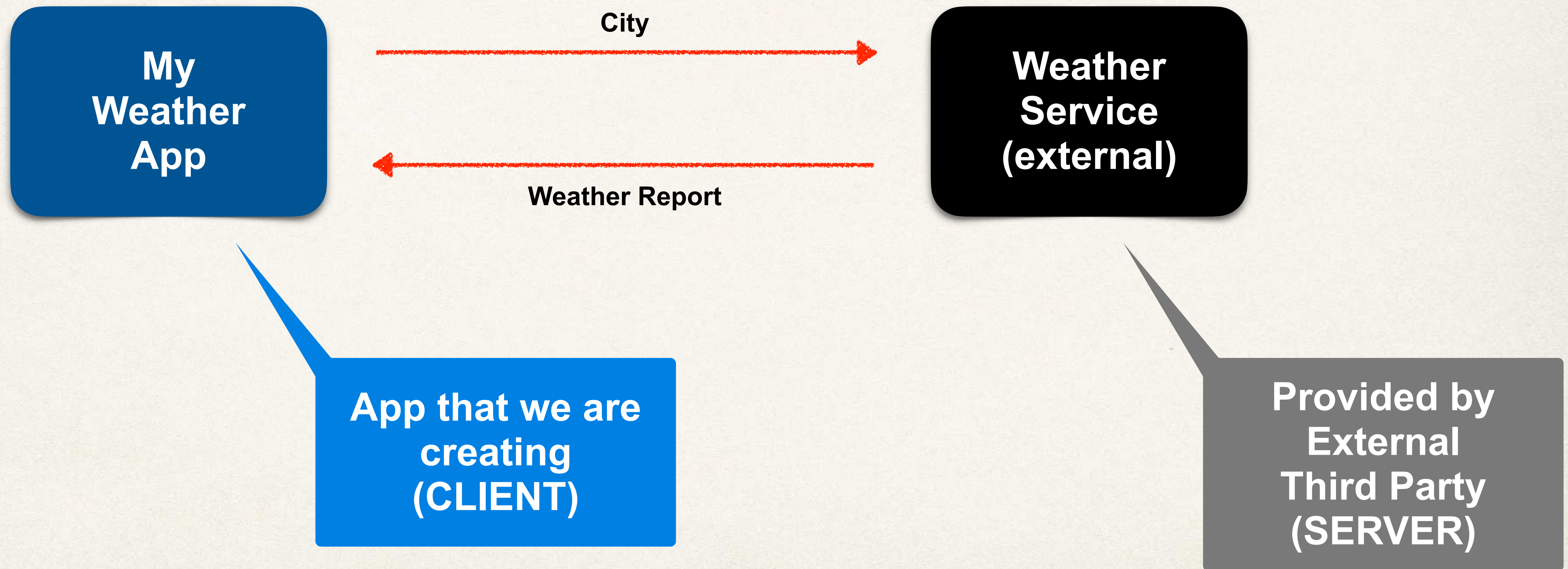
- Introduction to Spring REST development
- Not an A to Z reference ... for that you can see [Spring Reference Manual](#)

<https://projects.spring.io/spring-framework/>

Business Problem

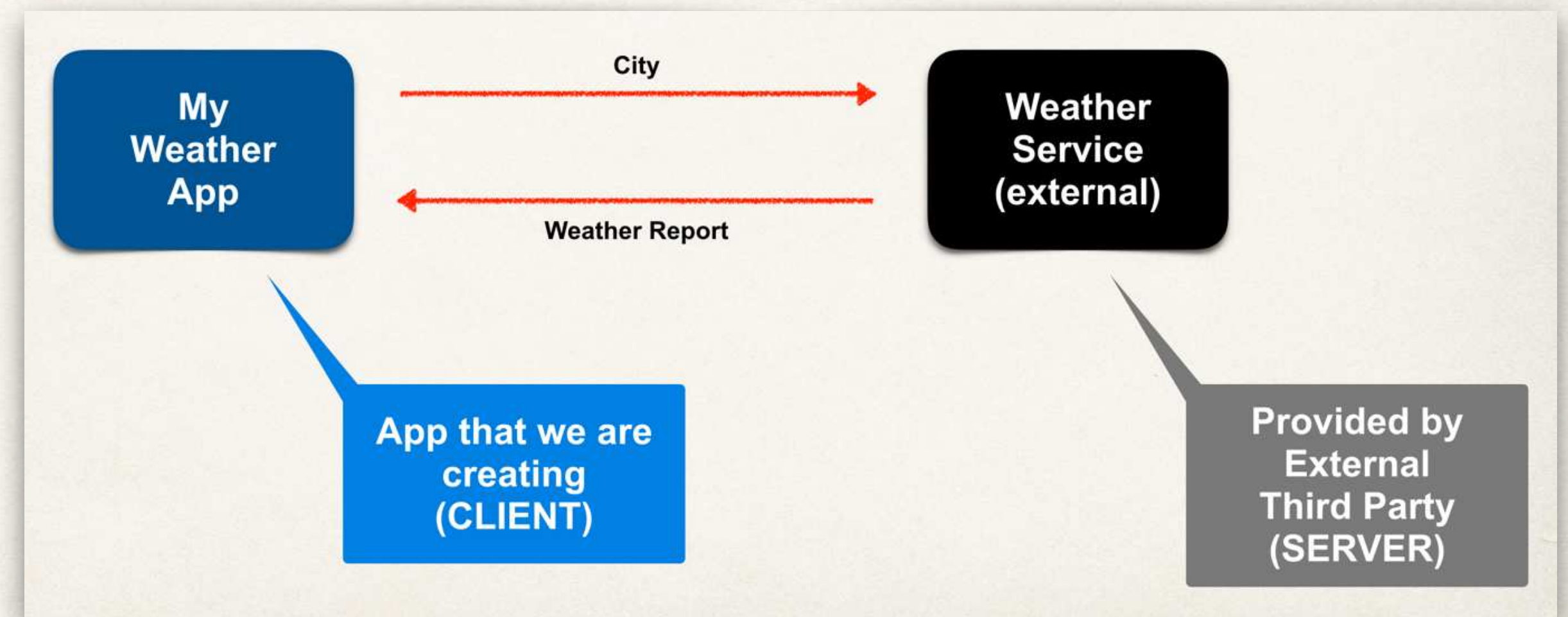
- Build a client app that provides the weather report for a city
- Need to get weather data from an external service

Application Architecture



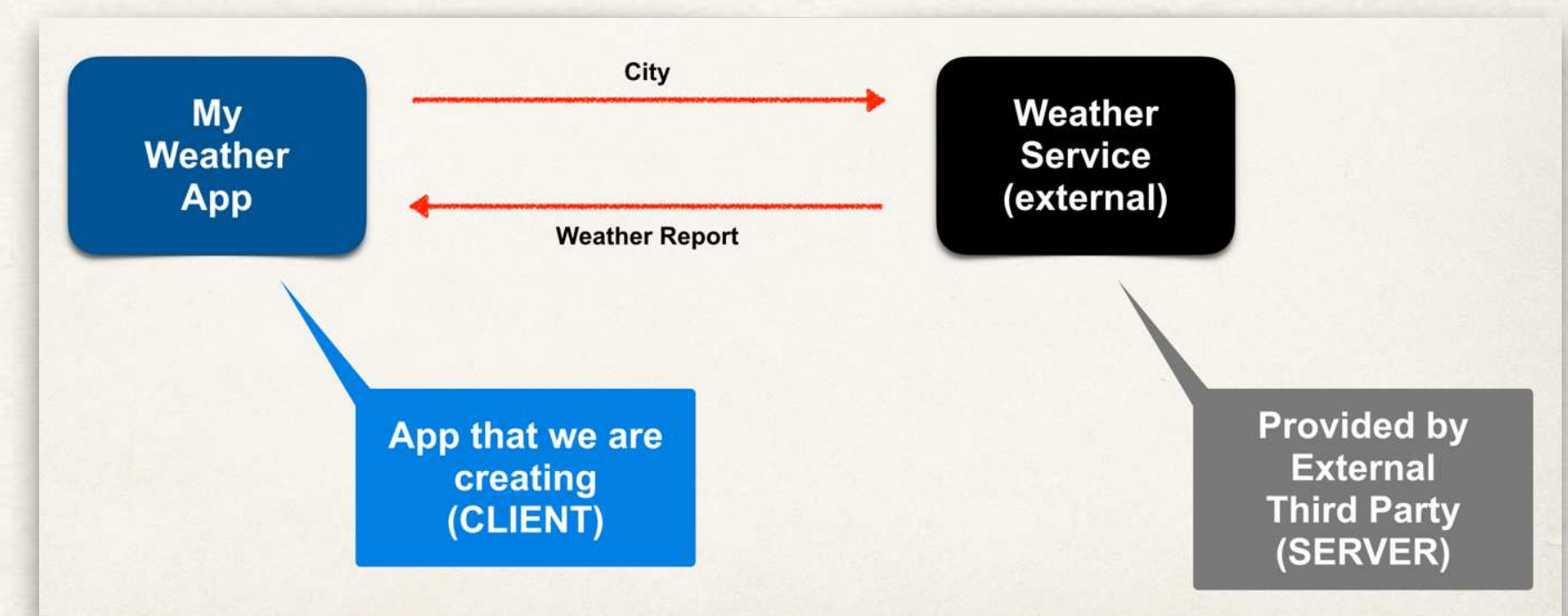
Questions

- How will we connect to the Weather Service?
- What programming language do we use?
- What is the data format?



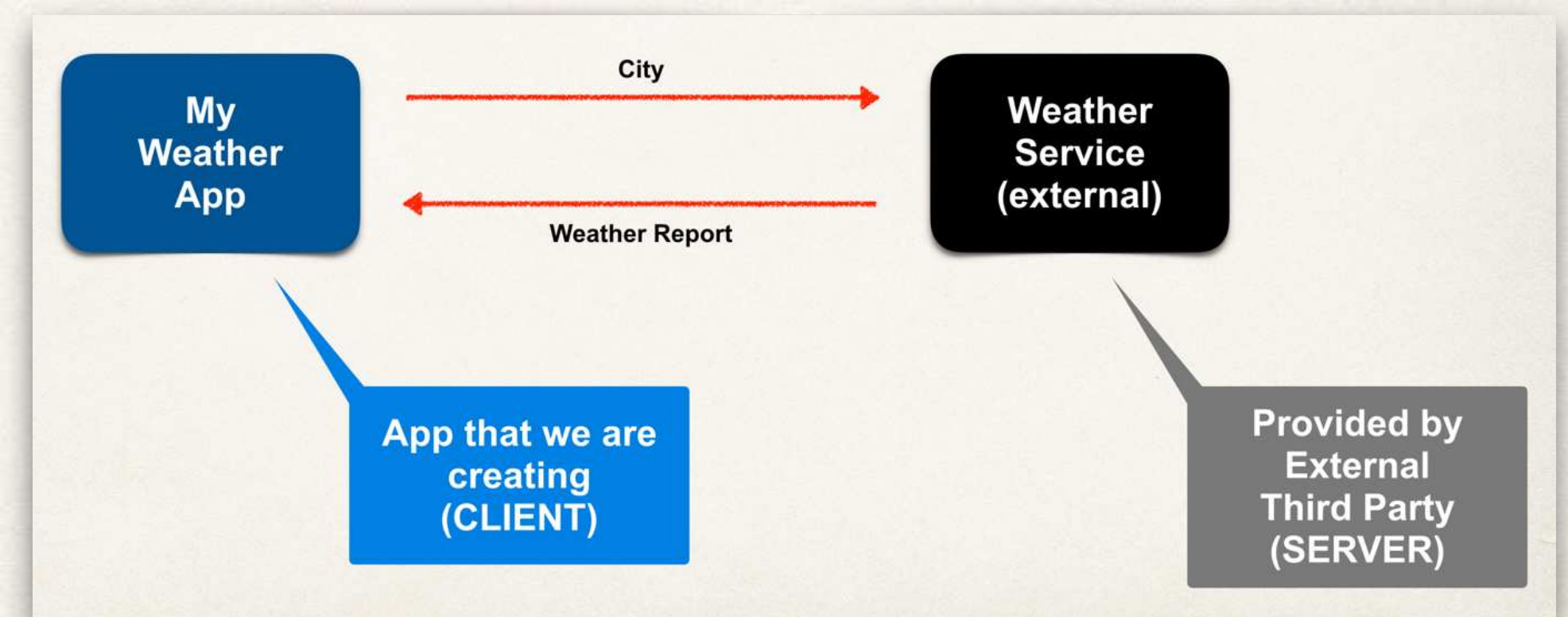
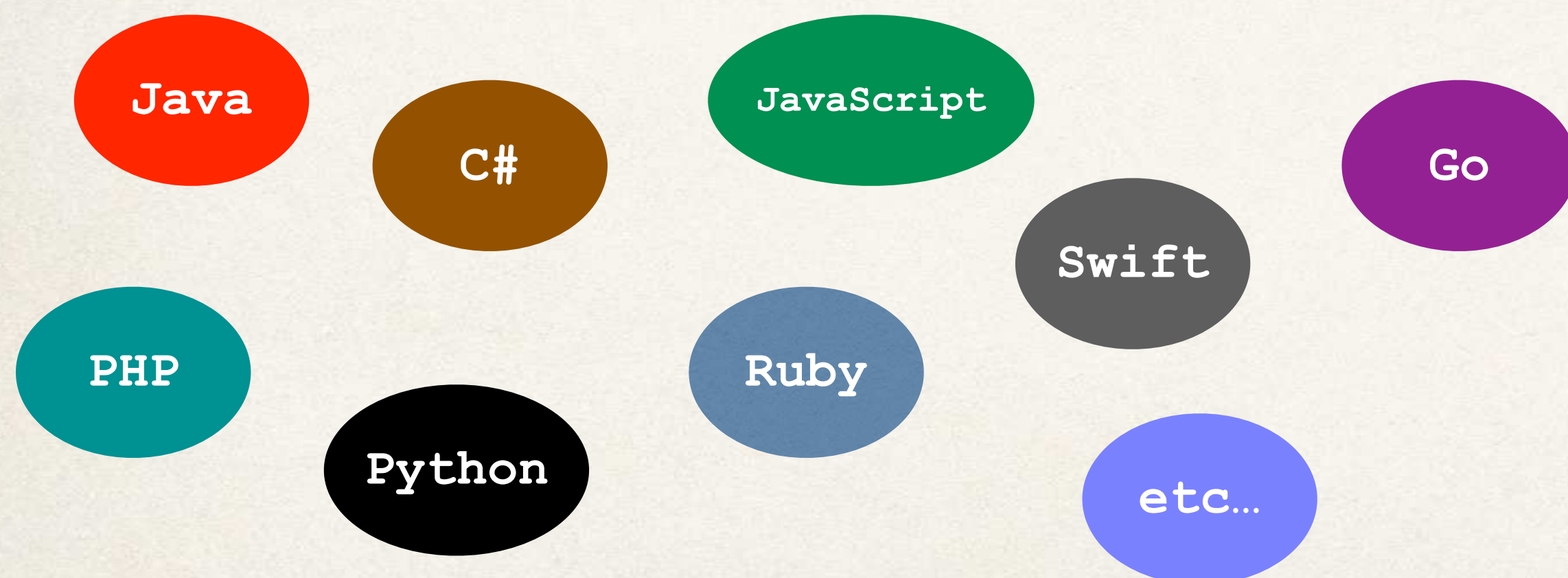
Answers

- How will we connect to the Weather Service?
- We can make REST API calls over HTTP
- REST: REpresentational State Transfer
- Lightweight approach for communicating between applications



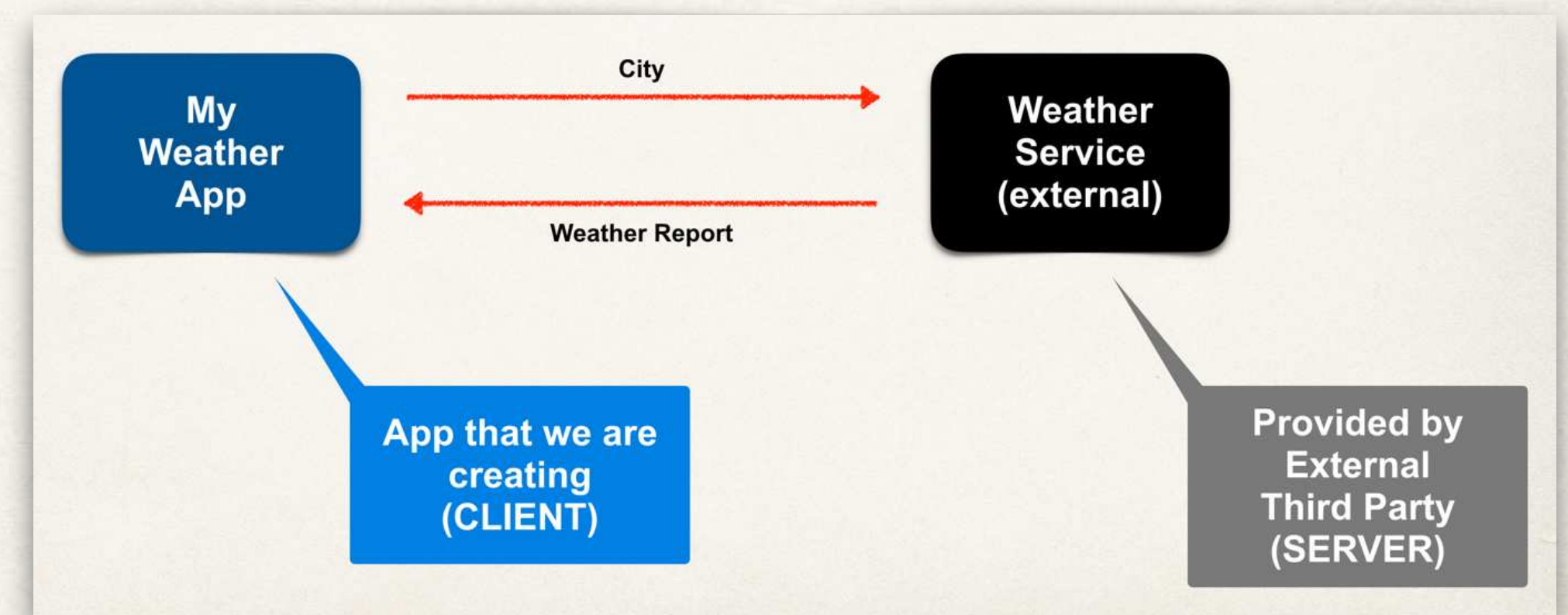
Answers

- What programming language do we use?
- REST is language independent
- The **client** application can use **ANY** programming language
- The **server** application can use **ANY** programming language



Answers

- What is the data format?
 - REST applications can use any data format
 - Commonly see XML and JSON
 - JSON is most popular and modern
 - JavaScript Object Notation

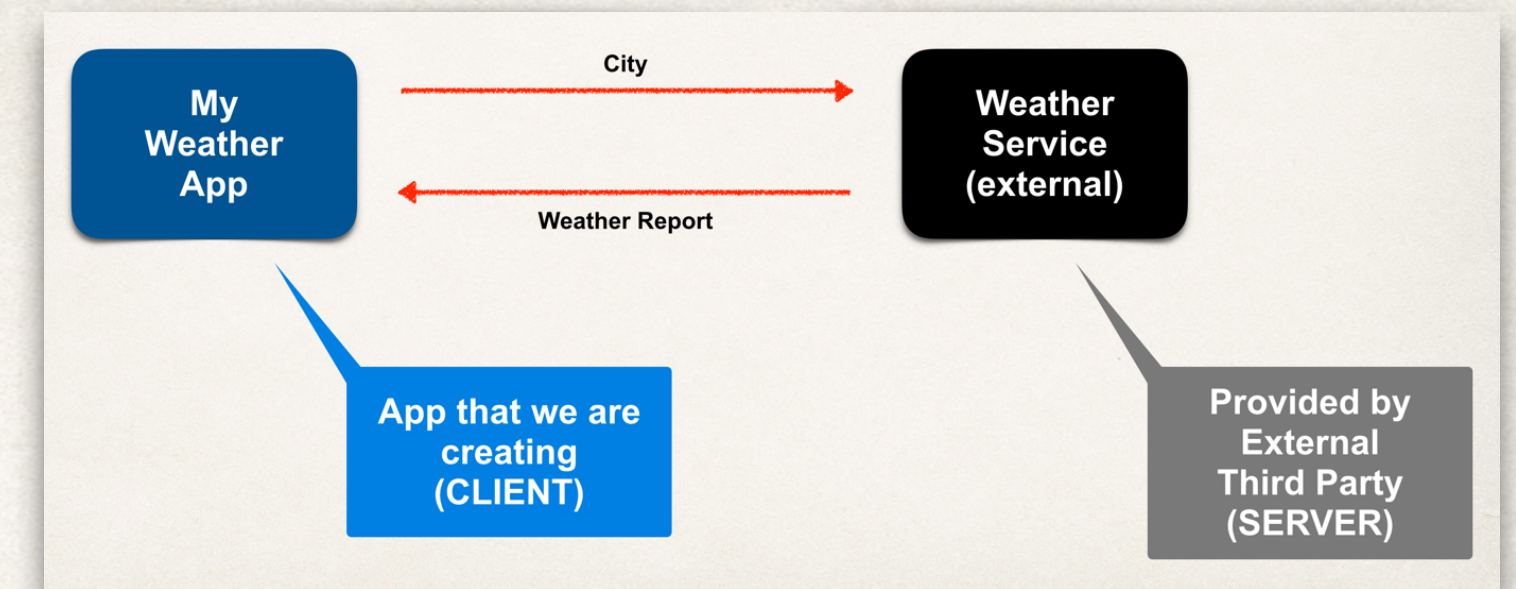


Possible Solution

- Use online Weather Service API provided by: openweathermap.org
- Provide weather data via an API
- Data is available in multiple formats: JSON, XML etc ...

Call Weather Service

- The API documentation gives us the following:
 - Pass in the latitude and longitude for your desired location



```
api.openweathermap.org/data/<apiVersion>/onecall?lat={theLatitude}&lon={theLongitude}
```

Pass the
latitude and longitude
of the city

Response - Weather Report

- The Weather Service responds with JSON



```
{  
  ...  
  "temp": xxx,  
  "feels_like": yyy,  
  "humidity": zzz,  
  ...  
}
```

Condensed
version

Multiple Client Apps

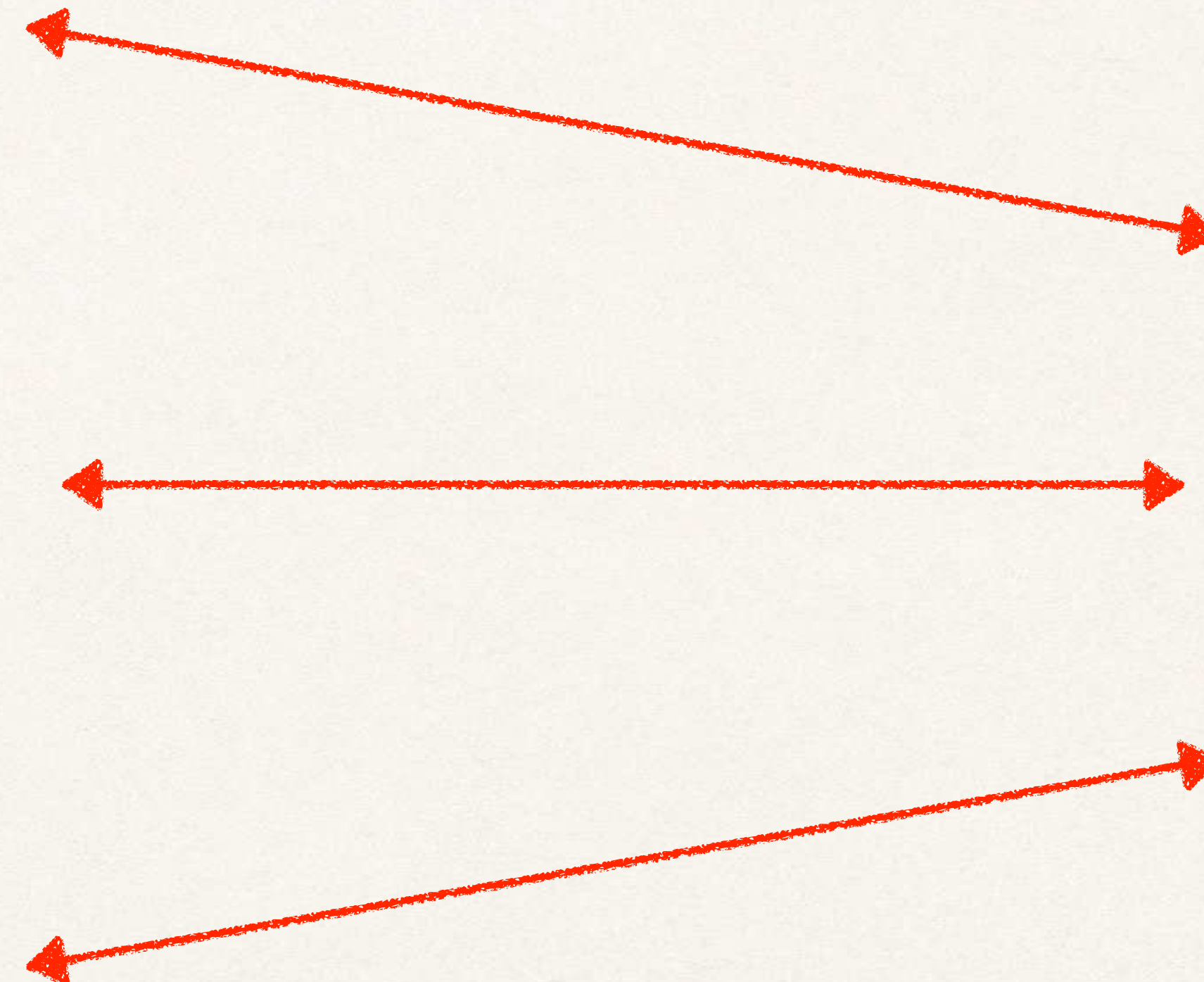
Remember:
REST calls can be made over HTTP
REST is language independent

My
Weather
Spring MVC

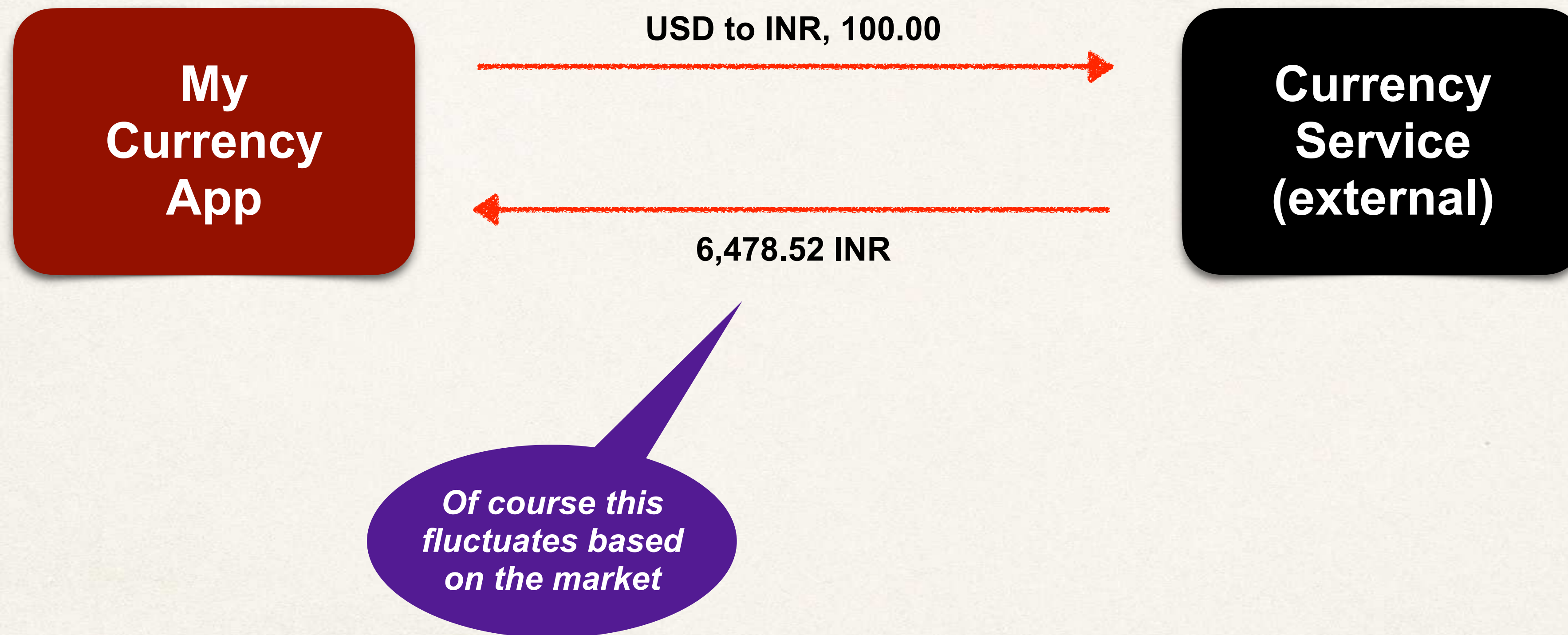
My
Weather
C# App

My
Weather
iPhone App

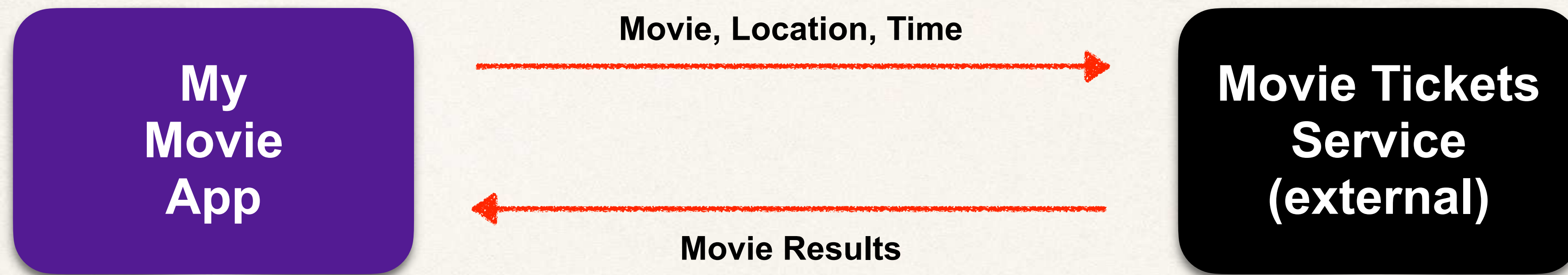
Weather
Service
(external)



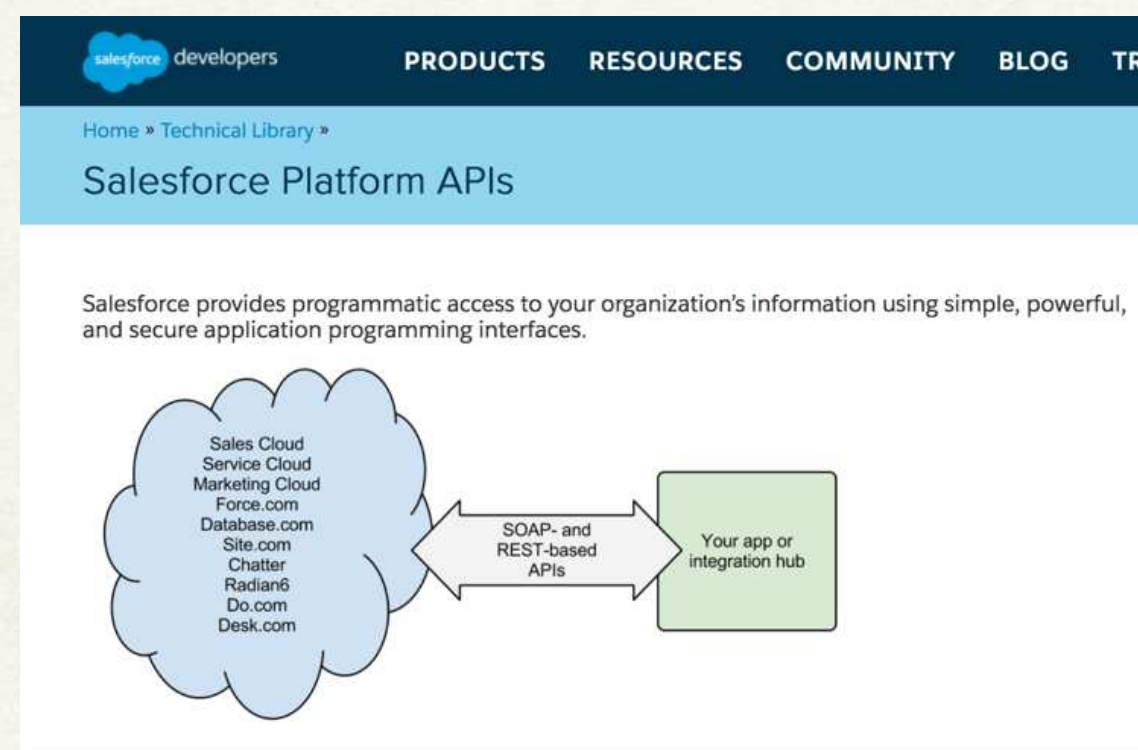
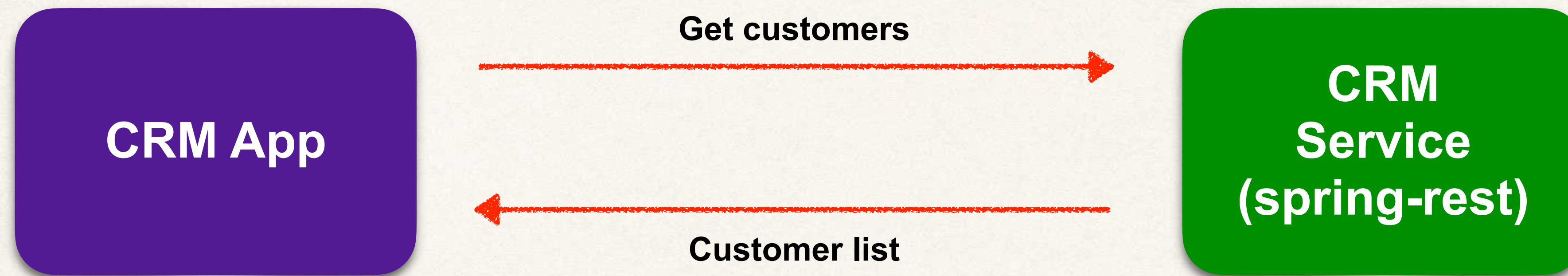
Currency Converter App



Movie Tickets App



Customer Relationship Manager (CRM) App



We will create
this code using
Spring REST
(SERVER)

What do we call it?

REST API

**REST
Web Services**

REST Services

RESTful API

**RESTful
Web Services**

RESTful Services

Generally, all mean the SAME thing

What do we call it?

REST API

**REST
Web Services**

REST Services

RESTful API

**RESTful
Web Services**

RESTful Services

Generally, all mean the SAME thing

JSON Basics - Syntax



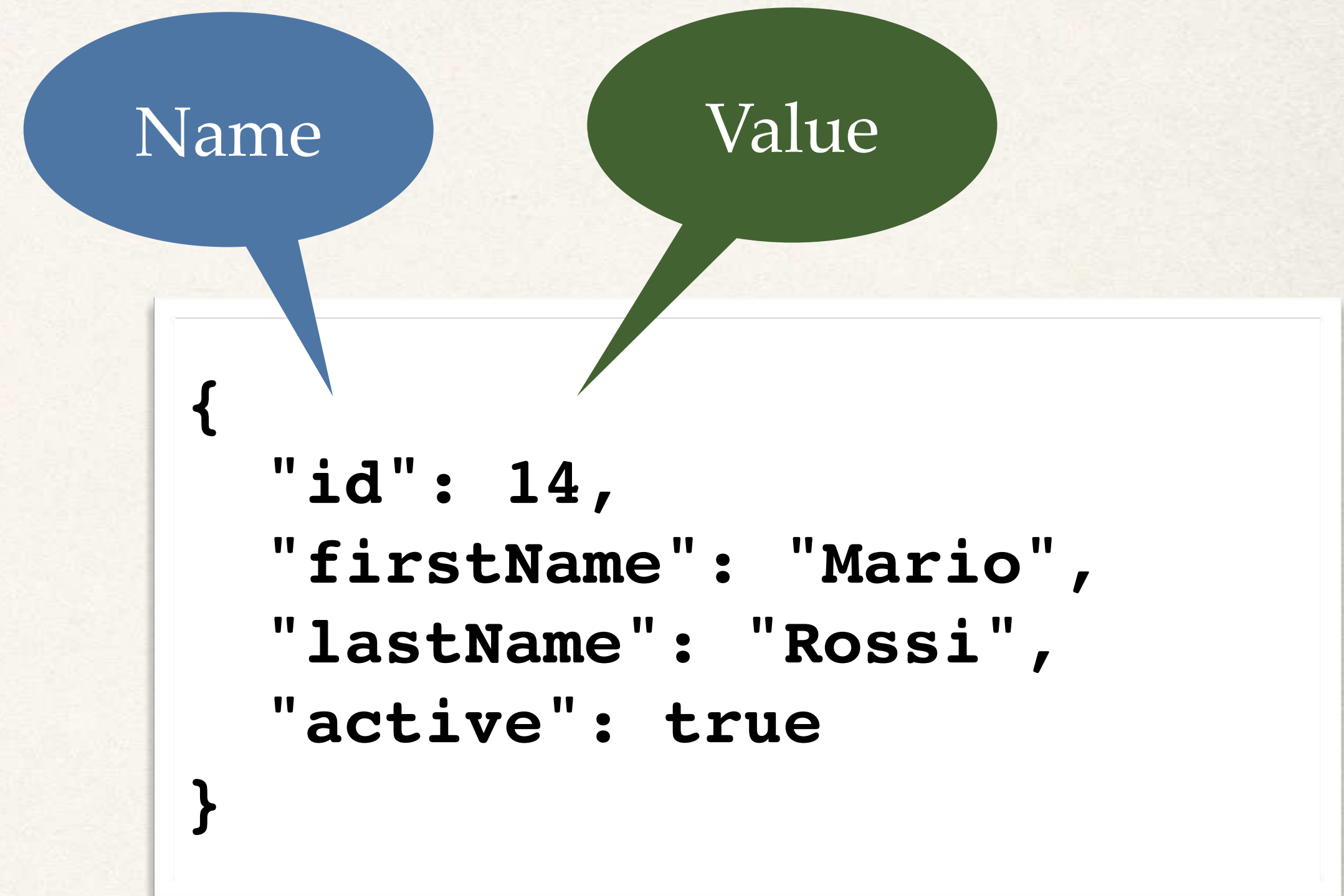
What is JSON?

- JavaScript Object Notation
- Lightweight data format for storing and exchanging data ... plain text
- Language independent ... not just for JavaScript
- Can use with any programming language: Java, C#, Python etc ...

**JSON is just
plain text
data**

Simple JSON Example

- Curley braces define objects in JSON
- Object members are name / value pairs
 - Delimited by colons
- Name is **always** in double-quotes



JSON Values

- Numbers: no quotes
- String: in double quotes
- Boolean: **true**, **false**
- Nested JSON object
- Array
- **null**



```
{  
  "id": 14,  
  "firstName": "Mario",  
  "lastName": "Rossi",  
  "active": true,  
  "courses" : null  
}
```

Nested JSON Objects

```
{  
  "id": 14,  
  "firstName": "Mario",  
  "lastName": "Rossi",  
  "active": true,  
  "address" : {  
    "street" : "100 Main St",  
    "city" : "Philadelphia",  
    "state" : "Pennsylvania",  
    "zip" : "19103",  
    "country" : "USA"  
  }  
}
```



JSON Arrays

```
{  
  "id": 14,  
  "firstName": "Mario",  
  "lastName": "Rossi",  
  "active": true,  
  "languages" : ["Java", "C#", "Python", "Javascript"]  
}
```



Array

REST HTTP Basics



REST over HTTP

- Most common use of REST is over HTTP
- Leverage HTTP methods for CRUD operations

HTTP Method	CRUD Operation
POST	<u>C</u> reate a new entity
GET	<u>R</u> ead a list of entities or single entity
PUT	<u>U</u> pdate an existing entity
DELETE	<u>D</u> eleate an existing entity

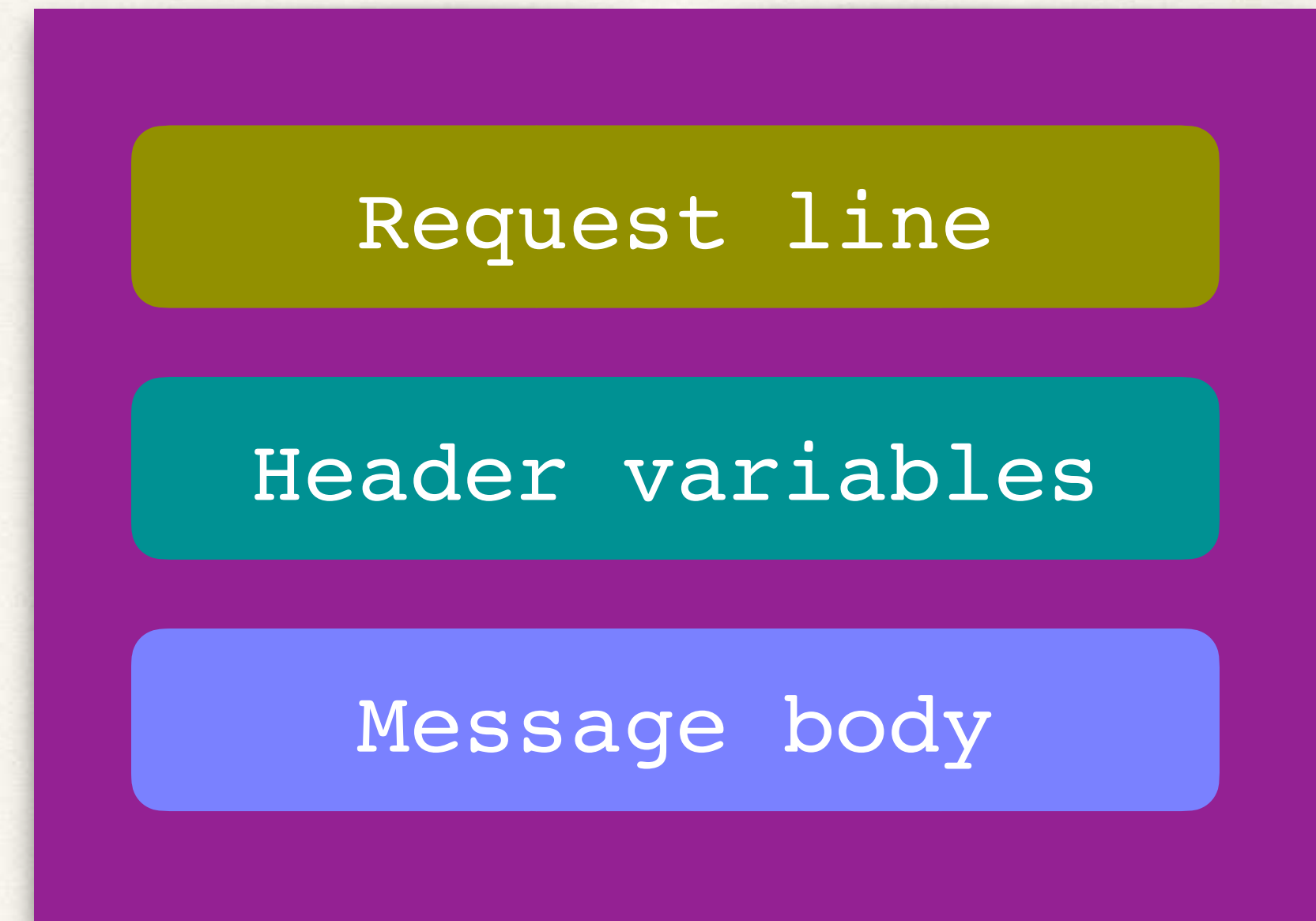
HTTP Messages



HTTP Request Message

- Request line: the HTTP command
- Header variables: request metadata
- Message body: contents of message

HTTP Request Message



HTTP Response Message

- Response line: server protocol and status code
- Header variables: response metadata
- Message body: contents of message

HTTP Response Message

Response line

Header variables

Message body

HTTP Response - Status Codes

Code Range	Description
100 - 199	Informational
200 - 299	Successful
300 - 399	Redirection
400 - 499	Client error
500 - 599	Server error

401 Authentication Required
404 File Not Found

500 Internal Server Error

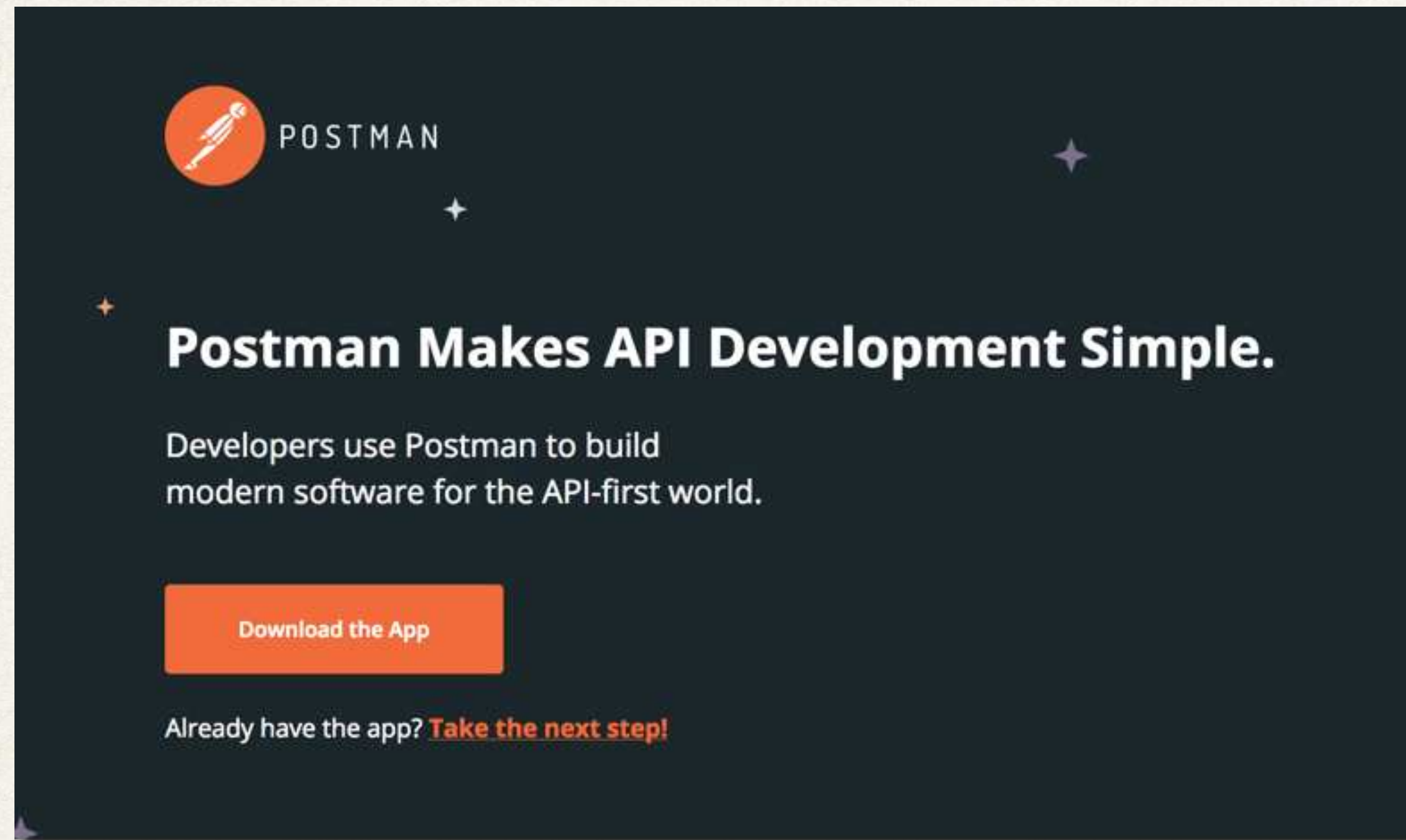
MIME Content Types

- The message format is described by MIME content type
 - Multipurpose Internet Mail-Extension
- Basic Syntax: `type/sub-type`
- Examples
 - `text/html`, `text/plain`
 - `application/json`, `application/xml`, ...

Client Tool

- We need a client tool
- Send HTTP requests to the REST Web Service / API
- Plenty of tools available: curl, Postman, etc ...

Postman



Free
developer
plan

www.getpostman.com

Install Postman Now

www.getpostman.com

Choose your platform:



Postman for Mac
for OS X Yosemite or later

Download



Postman for Windows
for Windows 7 or later

x64  Download



Postman for Linux

x64  Download

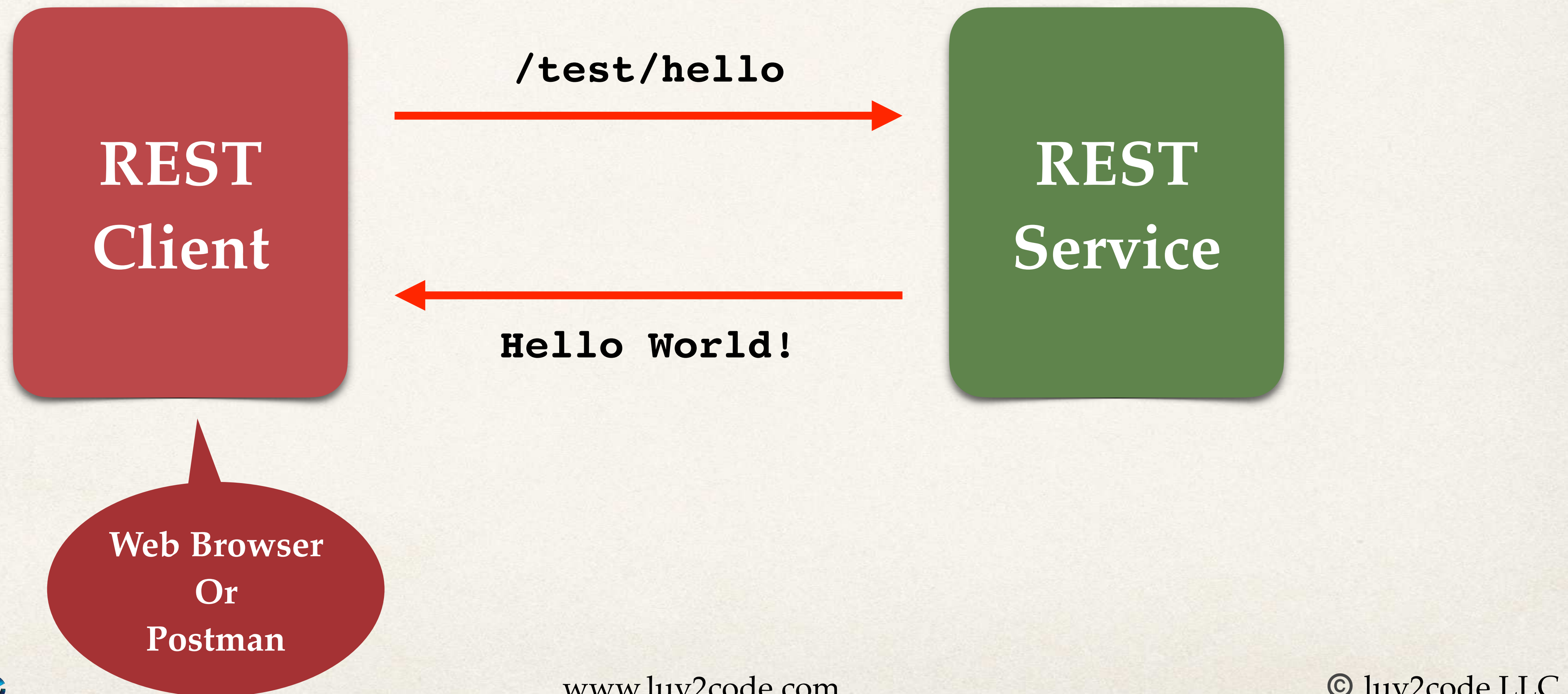
Postman Demo



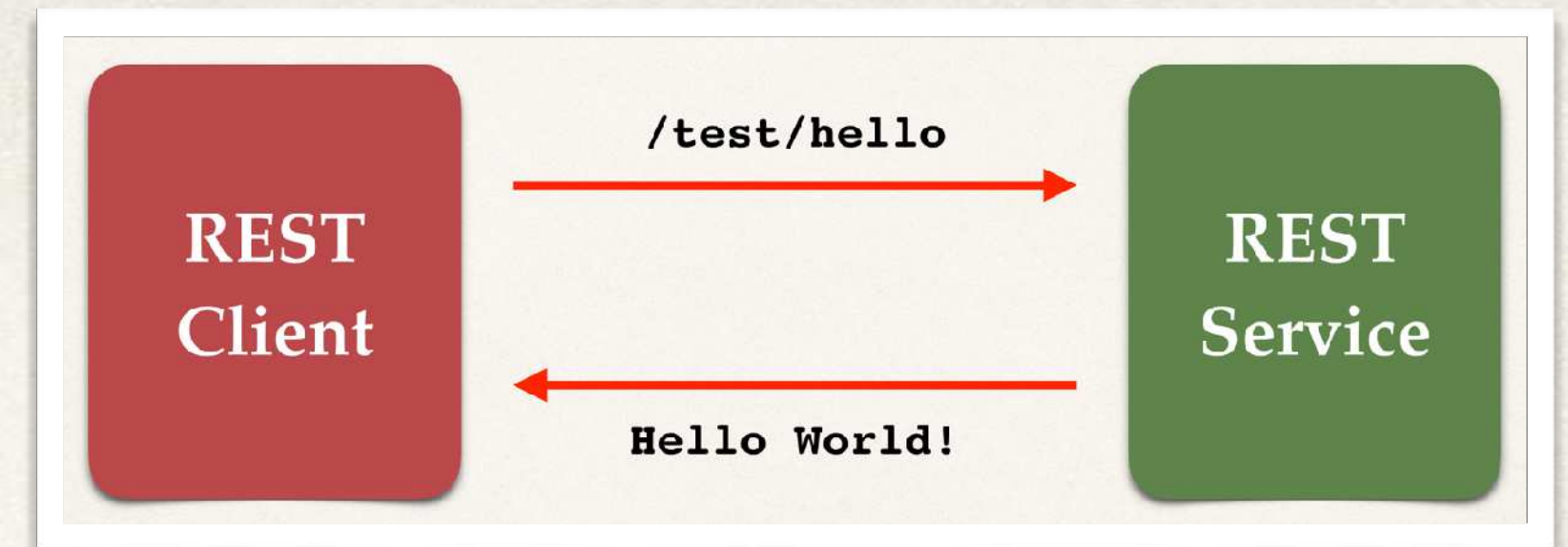
Spring REST Controller



Spring REST Hello World



Spring REST Controller



Adds REST support

```
@RestController
@RequestMapping("/test")
public class DemoRestController {

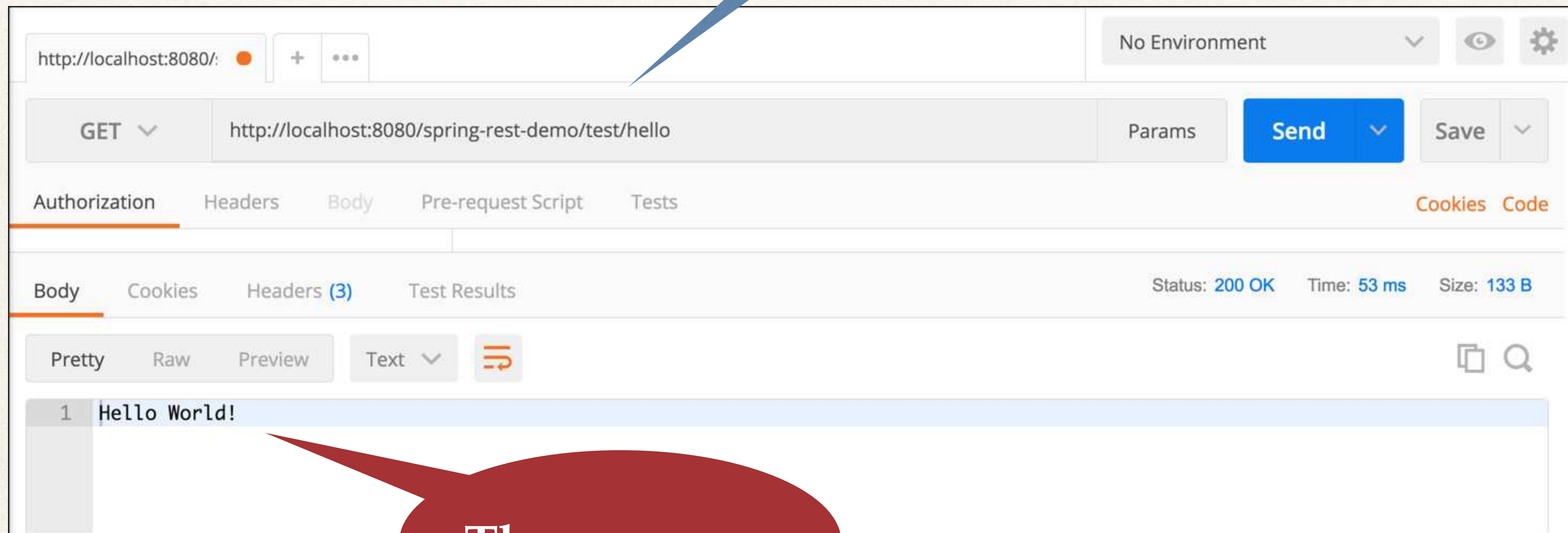
    @GetMapping("/hello")
    public String sayHello() {
        return "Hello World!";
    }
}
```

Access the REST endpoint at
/test/hello

Returns content to
client

Testing with REST Client - Postman

Access the REST endpoint at
`/test/hello`



The response

Testing with REST Client - Web Browser

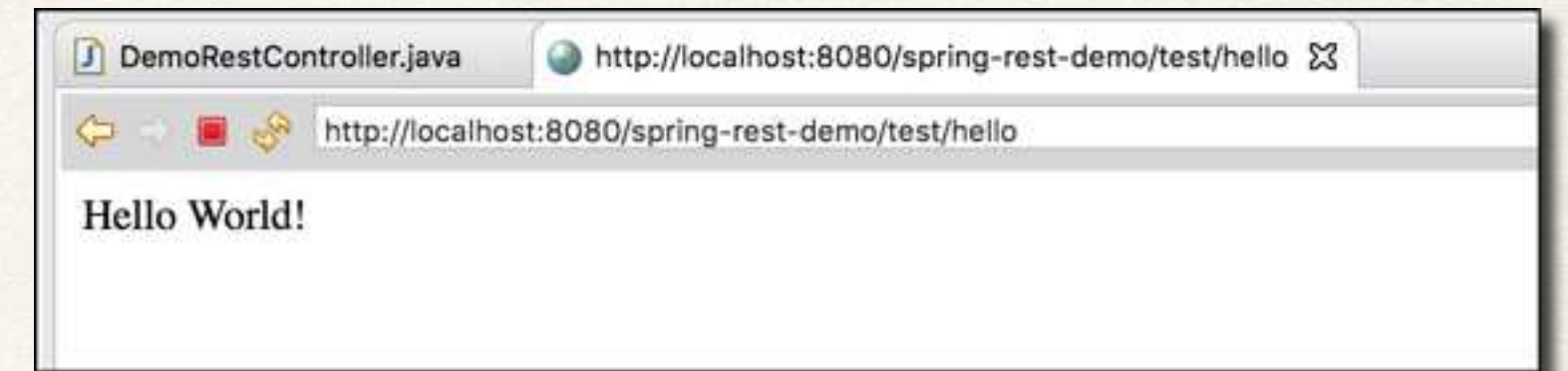


Access the REST endpoint at
`/test/hello`

The response

Web Browser vs Postman

- For simple REST testing for GET requests
 - Web Browser and Postman are similar
- However, for advanced REST testing: POST, PUT etc ...
 - Postman has much better support
 - POSTing JSON data, setting content type
 - Passing HTTP request headers, authentication etc ...



Spring REST Controller - Dev Process



Development Process

Step-By-Step

1. Add Maven dependency for Spring Boot Starter Web
2. Create Spring REST Service using @RestController

Step 1: Add Maven Dependency

File: pom.xml

```
<!-- Add Spring Boot Starter Web -->  
<dependency>  
  <groupId>org.springframework.boot</groupId>  
  <artifactId>spring-boot-starter-web</artifactId>  
</dependency>
```

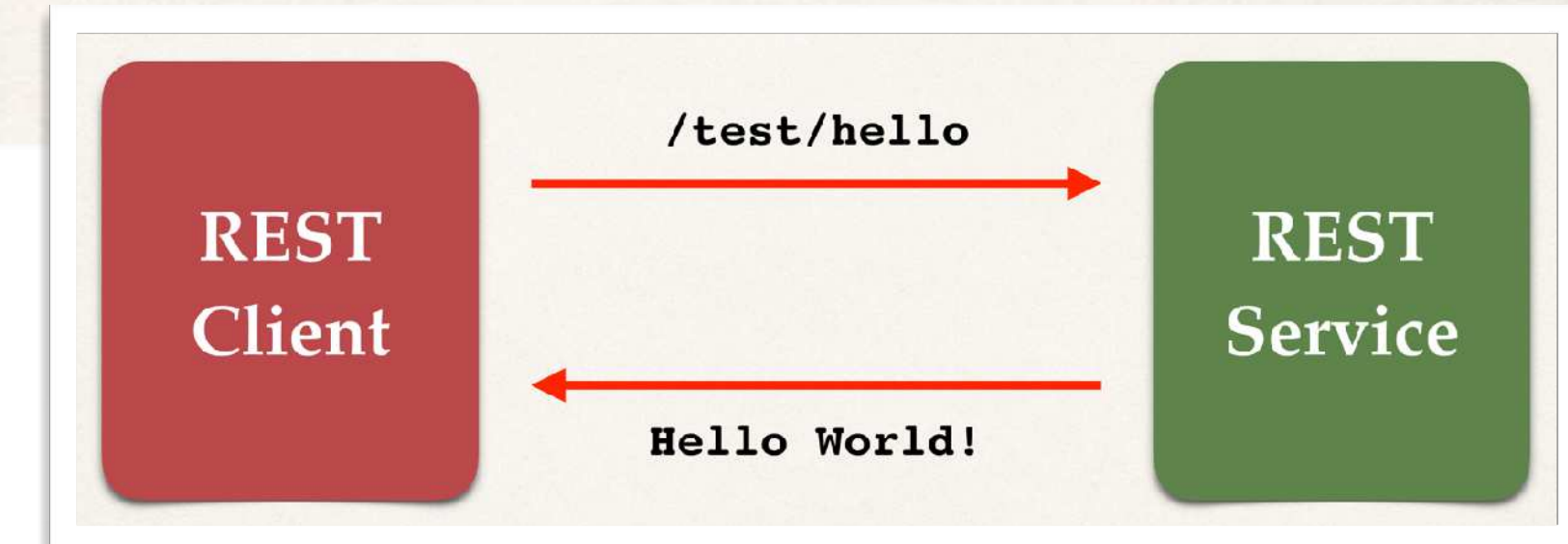
**At Spring Initializr website,
can also select the “Web” dependency**

Step 2: Create Spring REST Service

```
@RestController
@RequestMapping("/test")
public class DemoRestController {

    @GetMapping("/hello")
    public String sayHello() {
        return "Hello World!";
    }

}
```



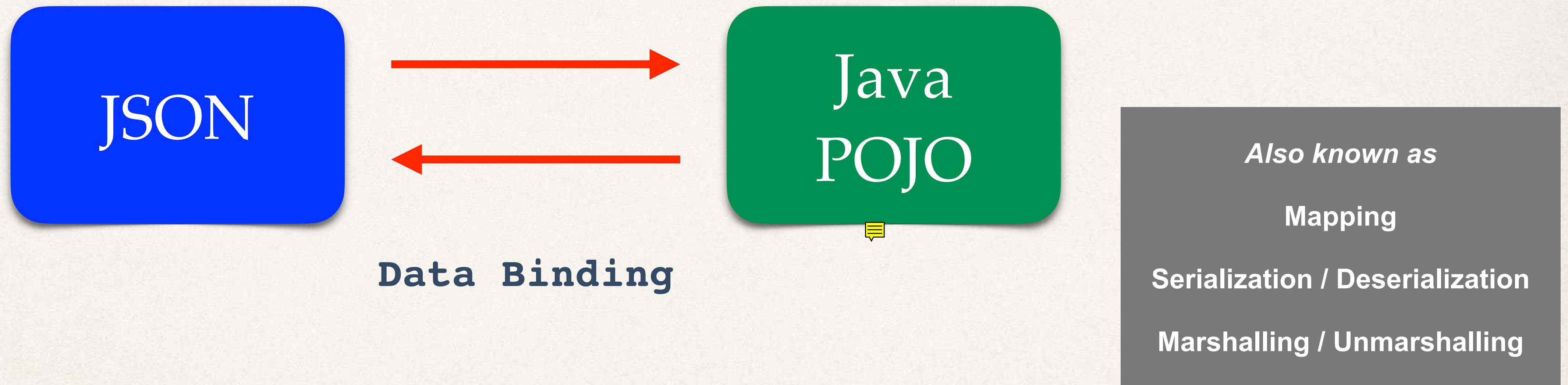
Handles HTTP GET requests

Java JSON Data Binding



Java JSON Data Binding

- Data binding is the process of converting JSON data to a Java POJO



JSON Data Binding with Jackson

- Spring uses the **Jackson Project** behind the scenes
- Jackson handles data binding between JSON and Java POJO
- Details on Jackson Project:

<https://github.com/FasterXML/jackson-databind>

Jackson Data Binding

- By default, Jackson will call appropriate getter / setter method



```
{  
  "id": 14,  
  "firstName": "Mario",  
  "lastName": "Rossi",  
  "active": true  
}
```



Java
POJO

Student

JSON to Java POJO

- Convert JSON to Java POJO ... call setter methods on POJO

```
{  
  "id": 14,  
  "firstName": "Mario",  
  "lastName": "Rossi",  
  "active": true  
}
```

*Call
setXXX
methods*



Jackson will do
this work

Java
POJO

Student

JSON to Java POJO

Note: Jackson calls the setXXX methods
It does NOT access internal private fields directly

- Convert JSON to Java POJO ... call setter methods on POJO

```
{  
  "id": 14,  
  "firstName": "Mario",  
  "lastName": "Rossi",  
  "active": true  
}
```

Call setId(...)

Call setFirstName(...)

Call setLastName(...)

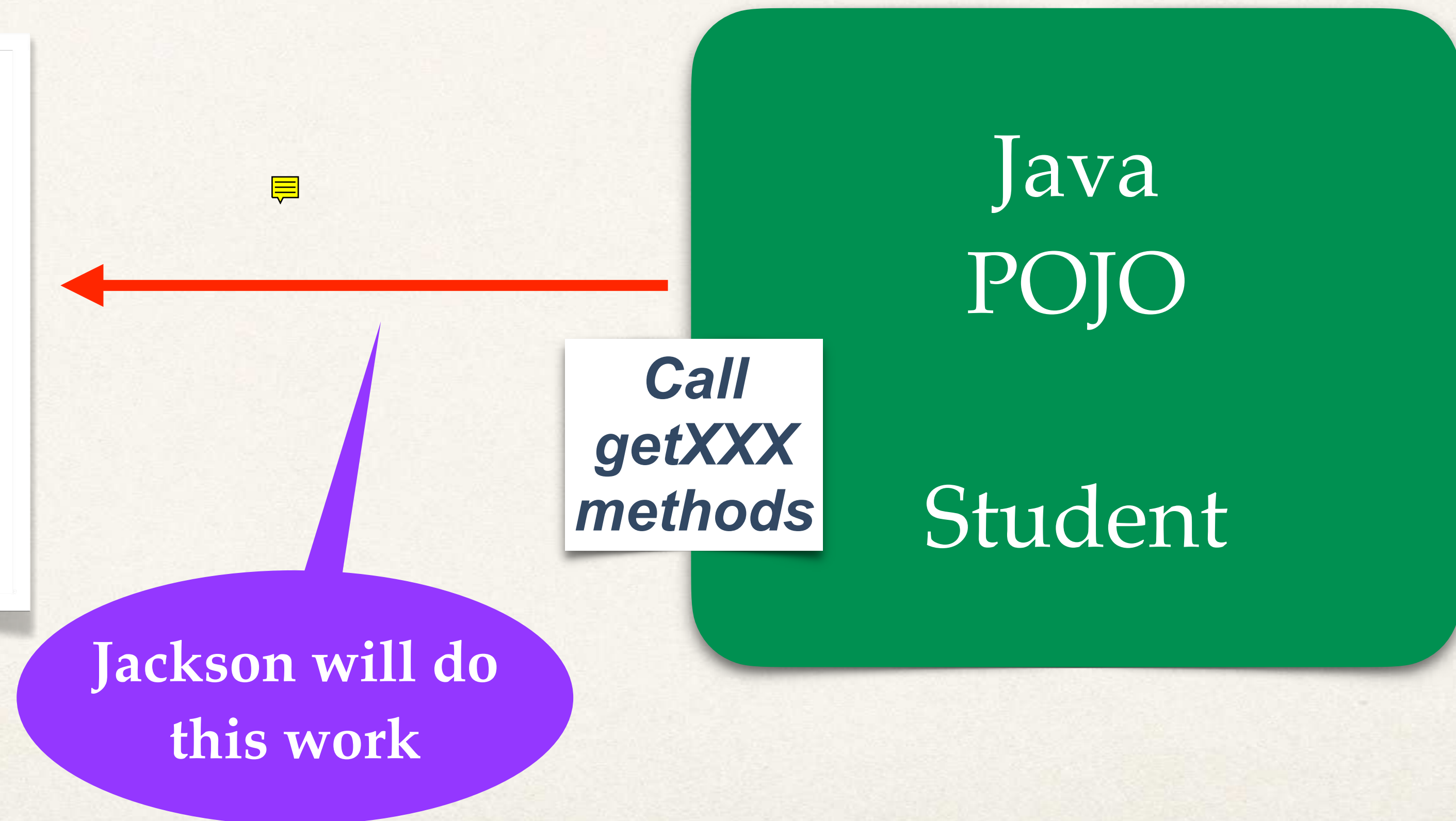
Jackson will do
this work

```
public class Student {  
  
    private int id;  
    private String firstName;  
    private String lastName;  
    private boolean active;  
  
    public void setId(int id) {  
        this.id = id;  
    }  
  
    public void setFirstName(String firstName) {  
        this.firstName = firstName;  
    }  
  
    public void setLastName(String lastName) {  
        this.lastName = lastName;  
    }  
  
    public void setActive(boolean active) {  
        this.active = active;  
    }  
  
    // getter methods  
}
```

Java POJO to JSON

- Now, let's go the other direction
- Convert Java POJO to JSON ... call getter methods on POJO

```
{  
  "id": 14,  
  "firstName": "Mario",  
  "lastName": "Rossi",  
  "active": true  
}
```



Spring and Jackson Support

- When building Spring REST applications
- Spring will automatically handle Jackson Integration
- JSON data being passed to REST controller is converted to POJO
- Java object being returned from REST controller is converted to JSON

**Happens
automatically
behind the scenes**

Spring REST Service - Students



Create a New Service

- Return a list of students

GET

/api/students

Returns a list of students

Spring REST Service

We will write
this code

REST
Client

`/api/students`

REST
Service

Web Browser
Or
Postman

```
[  
  {  
    "firstName": "Poornima",  
    "lastName": "Patel"  
  },  
  {  
    "firstName": "Mario",  
    "lastName": "Rossi"  
  },  
  {  
    "firstName": "Mary",  
    "lastName": "Smith"  
  }  
]
```

Convert Java POJO to JSON

- Our REST Service will return **List<Student>**
- Need to convert **List<Student>** to JSON
- Jackson can help us out with this ...

Spring Boot and Jackson Support

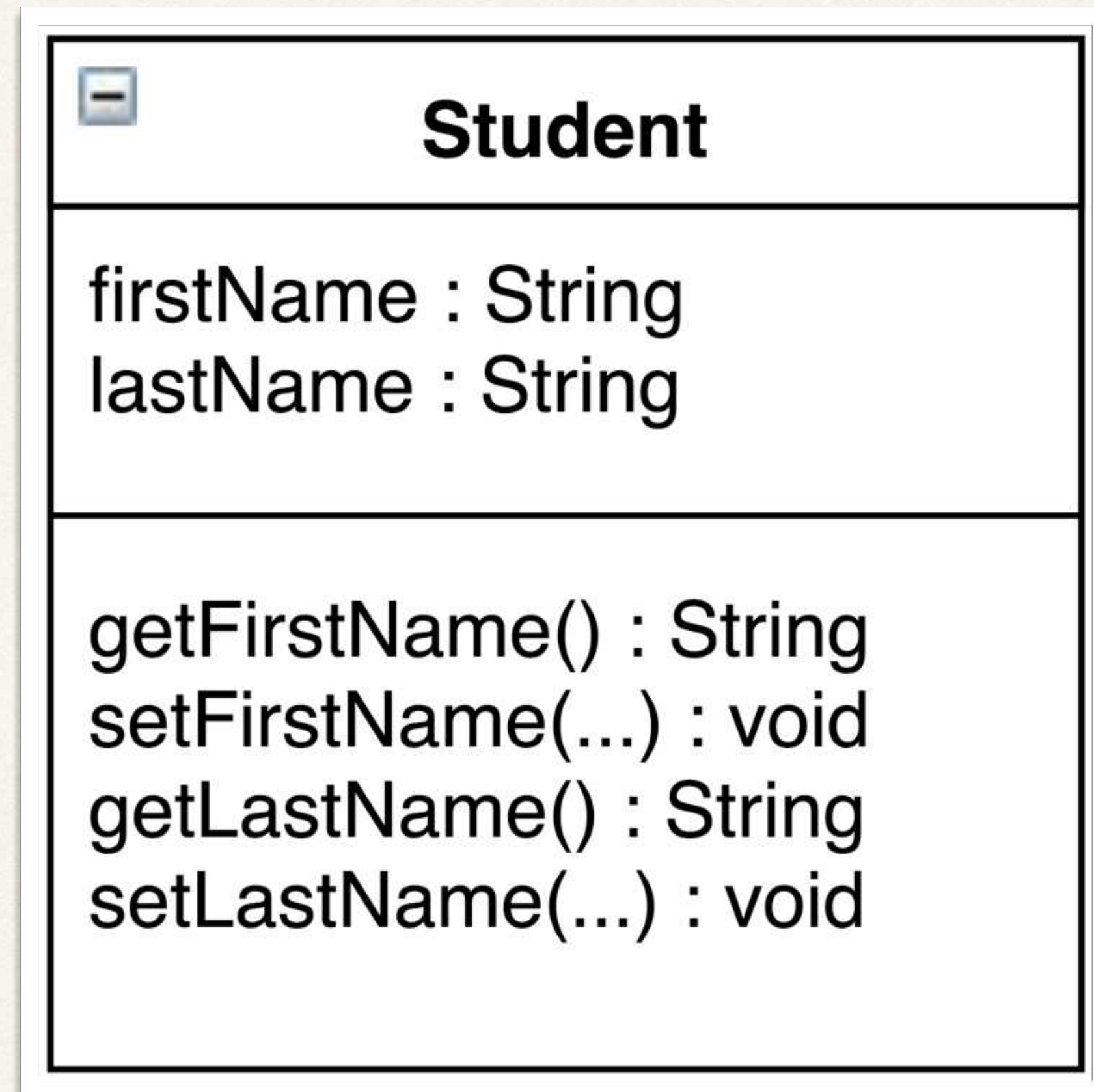
**Happens
automatically
behind the scenes**

- Spring Boot will automatically handle **Jackson** integration
- JSON data being passed to REST controller is converted to Java POJO
- Java POJO being returned from REST controller is converted to JSON

Student POJO (class)

Java
POJO

Student



Jackson Data Binding

- Jackson will call appropriate getter / setter method

Remember

```
{  
  "id": 14,  
  "firstName": "Mario",  
  "lastName": "Rossi",  
  "active": true  
}
```



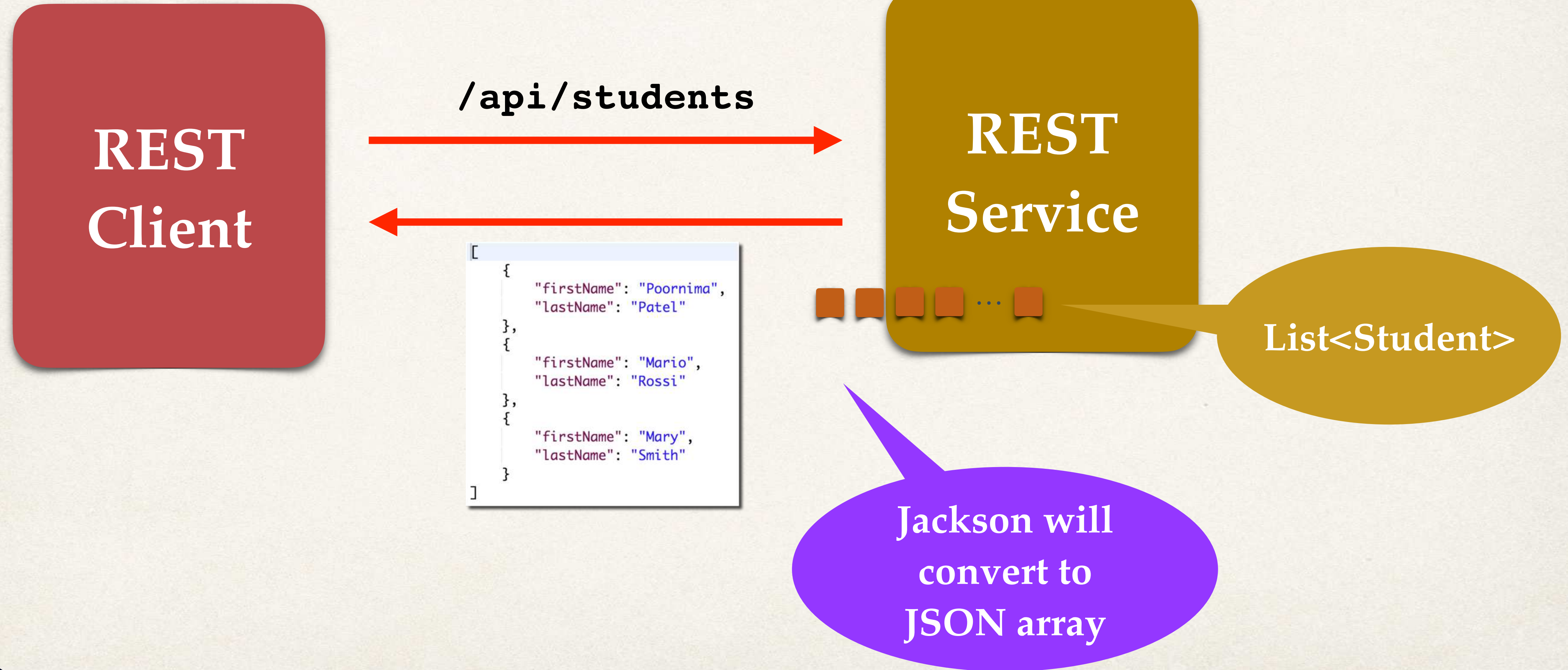
Jackson will do
this work

Java
POJO

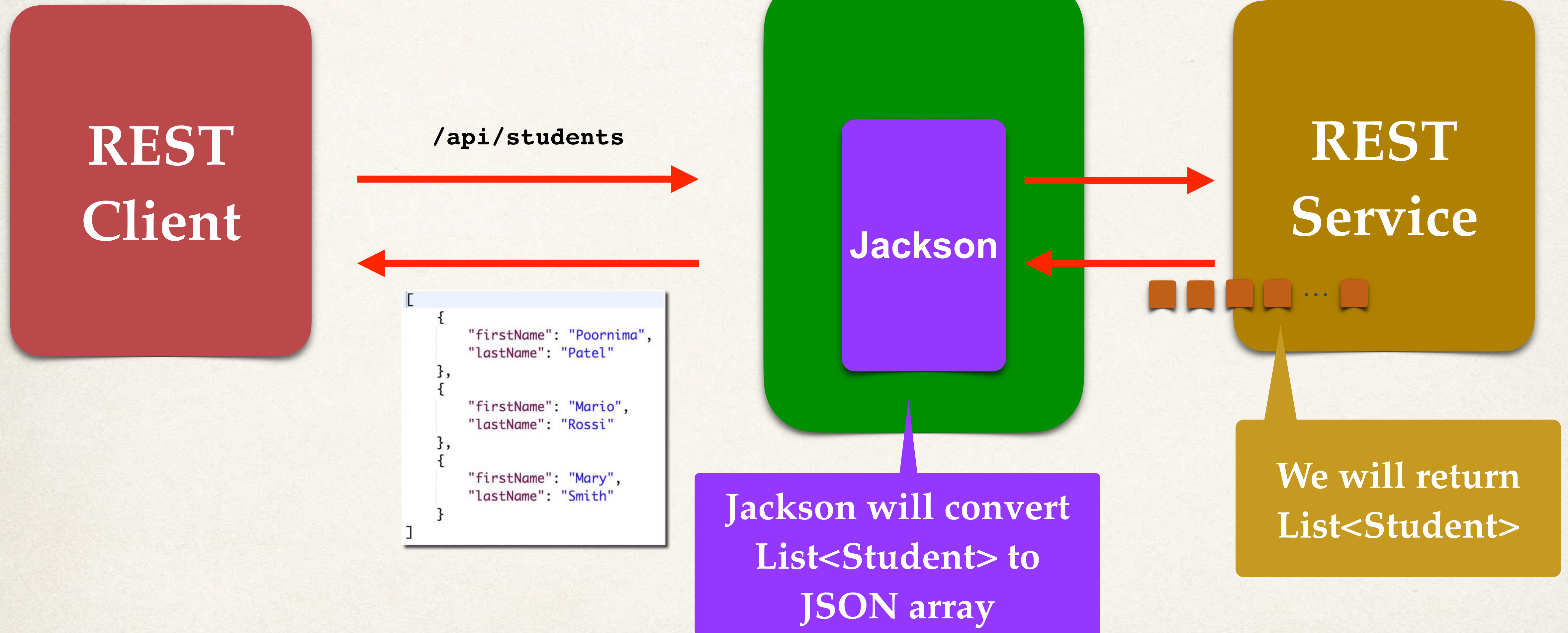
Student

Student
firstName : String lastName : String
getFirstName() : String setFirstName(...) : void getLastName() : String setLastName(...) : void

Spring REST Service



Behind the scenes



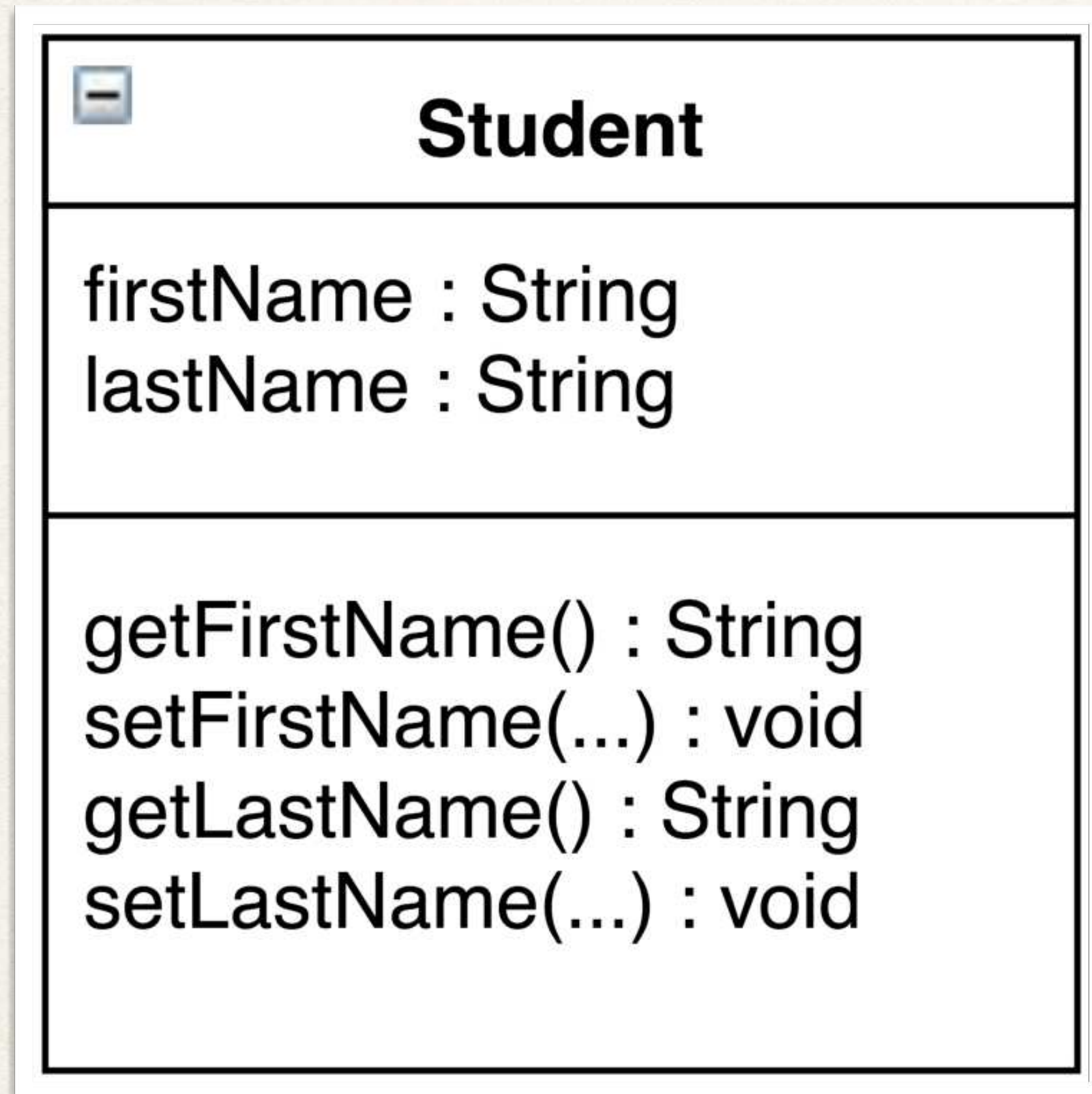
Development Process

Step-By-Step

1. Create Java POJO class for **Student**

2. Create Spring REST Service using **@RestController**

Step 1: Create Java POJO class for Student



File: Student.java

```
public class Student {

    private String firstName;
    private String lastName;

    public Student() {

    }

    public Student(String firstName, String lastName) {
        this.firstName = firstName;
        this.lastName = lastName;
    }

    public String getFirstName() {
        return firstName;
    }

    public void setFirstName(String firstName) {
        this.firstName = firstName;
    }

    public String getLastName() {
        return lastName;
    }

    public void setLastName(String lastName) {
        this.lastName = lastName;
    }

}
```

Fields
Constructors
Getter/Setters

Step 2: Create @RestController

File: StudentRestController.java

```
@RestController
@RequestMapping("/api")
public class StudentRestController {

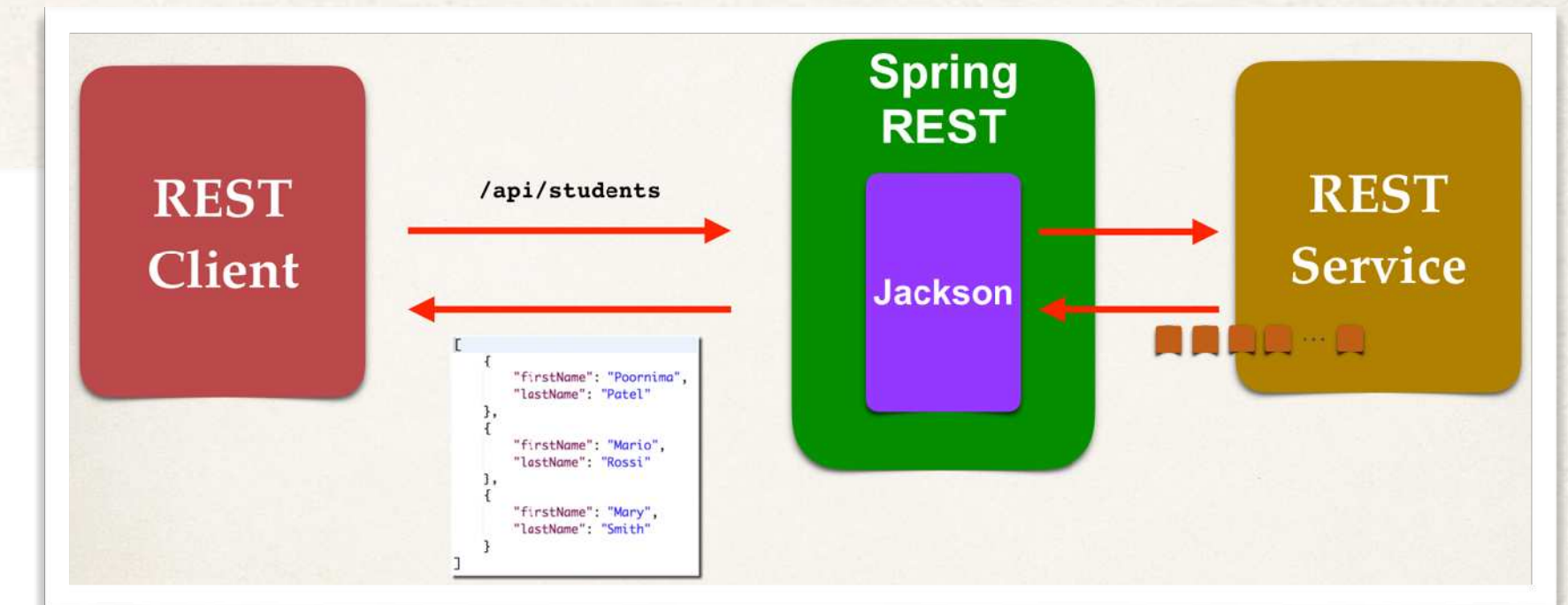
    // define endpoint for "/students" - return list of students

    @GetMapping("/students")
    public List<Student> getStudents() {

        List<Student> theStudents = new ArrayList<>();

        theStudents.add(new Student("Poornima", "Patel"));
        theStudents.add(new Student("Mario", "Rossi"));
        theStudents.add(new Student("Mary", "Smith"));

        return theStudents;
    }
}
```



We'll hard code
for now ...
can add DB later ...

Jackson will convert
List<Student> to
JSON array

Spring REST - Path Variables



Path Variables

- Retrieve a single student by id

GET

`/api/students/{studentId}` *Retrieve a single student*

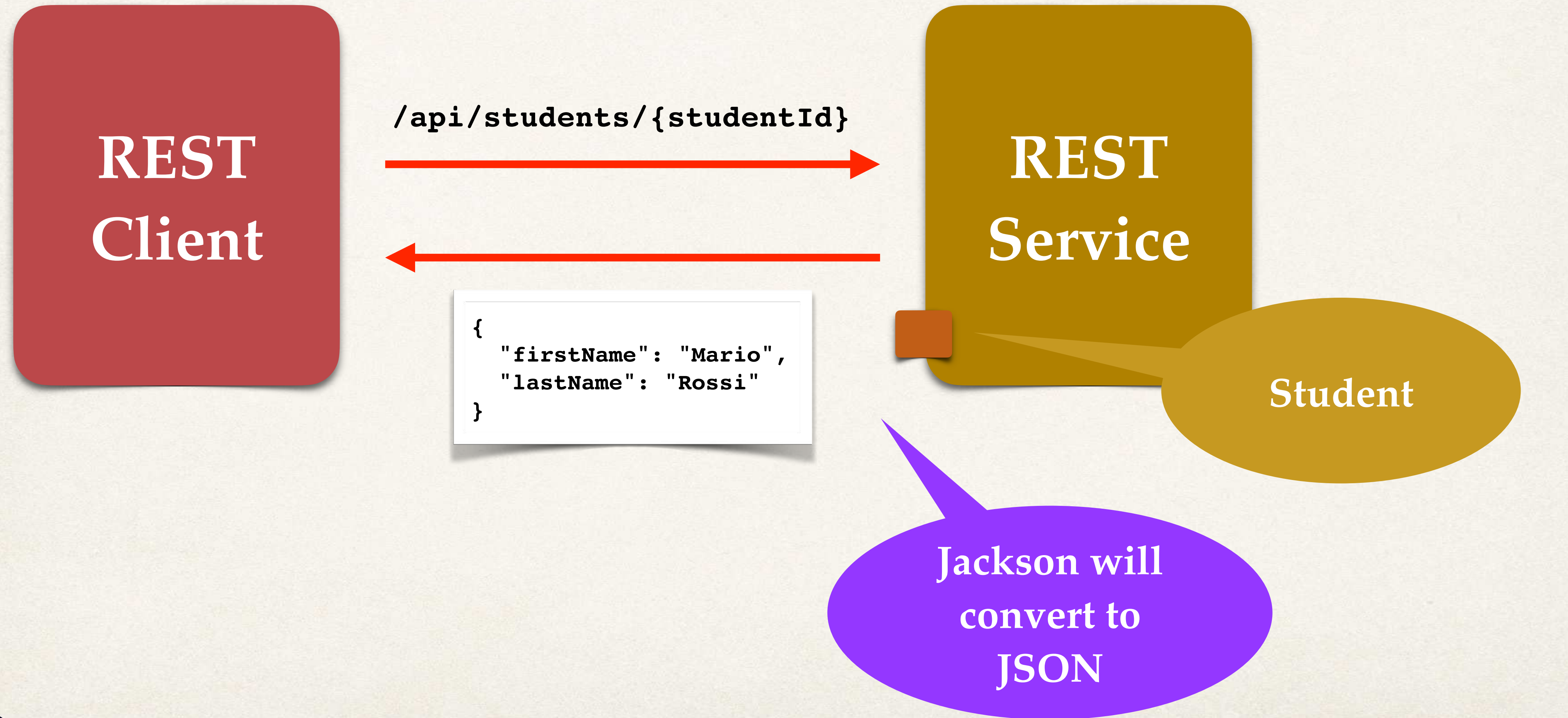
`/api/students/0`

`/api/students/1`

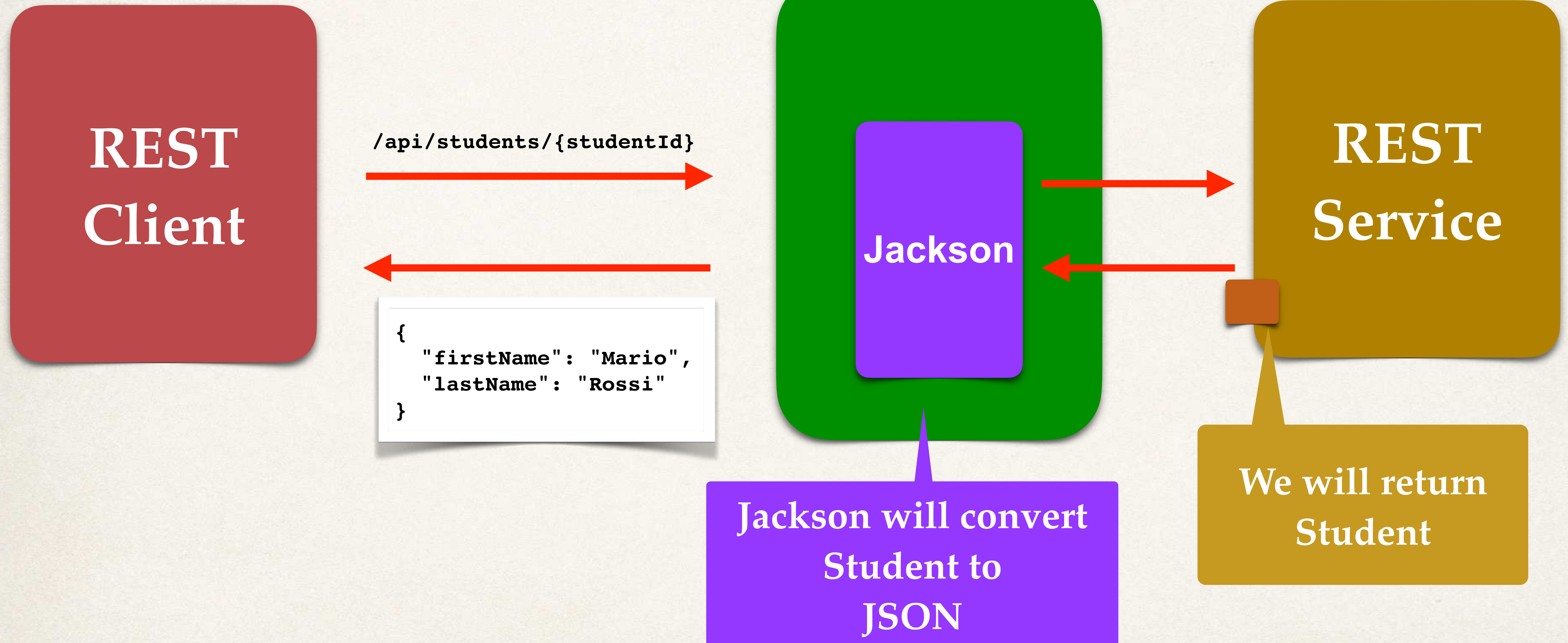
`/api/students/2`

Known as a
"path variable"

Spring REST Service



Behind the scenes



Development Process

Step-By-Step

1. Add request mapping to Spring REST Service
 - ◉ Bind path variable to method parameter using `@PathVariable`

Step 1: Add Request Mapping

File: StudentRestController.java

```
@RestController
@RequestMapping("/api")
public class StudentRestController {

    // define endpoint for "/students/{studentId}" - return student at index

    @GetMapping("/students/{studentId}")
    public Student getStudent(@PathVariable int studentId) {

        List<Student> theStudents = new ArrayList<>();

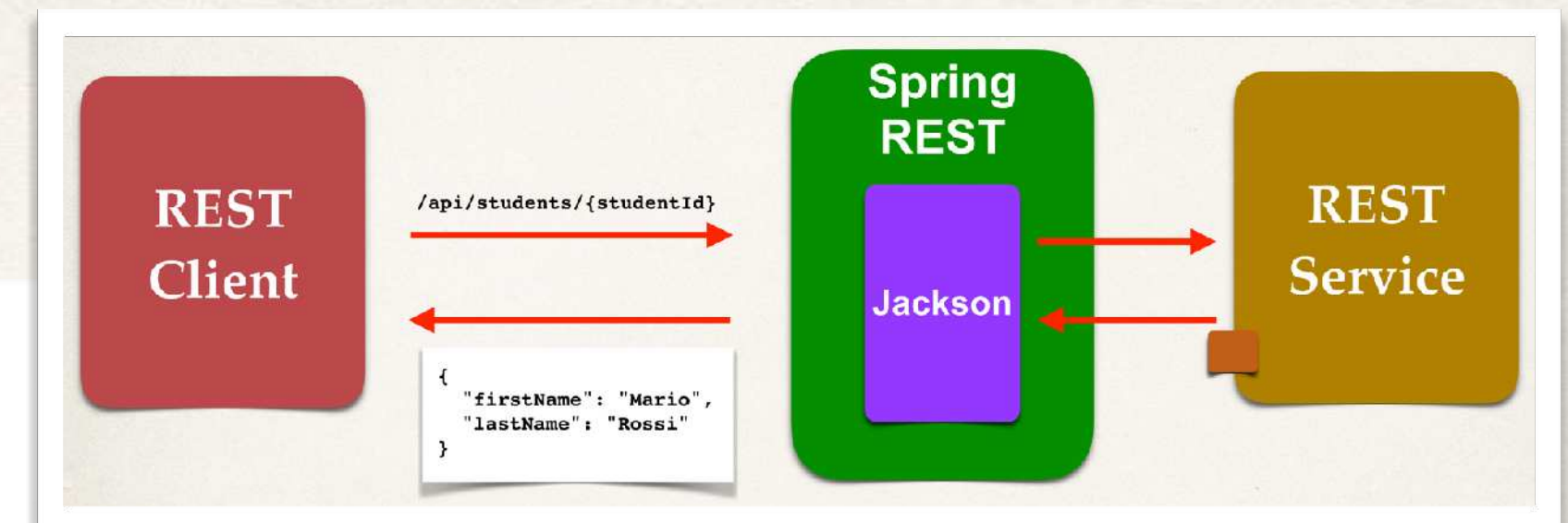
        // populate theStudents
        ...

        return theStudents.get(studentId);
    }
}
```

Keep it simple,
just index into the list
We'll do fancy DB stuff later

Bind the path variable
(by default, must match)

Jackson will convert
Student to JSON



Spring REST - Exception Handling



Remember our problem?

- Bad student id of 9999 ...

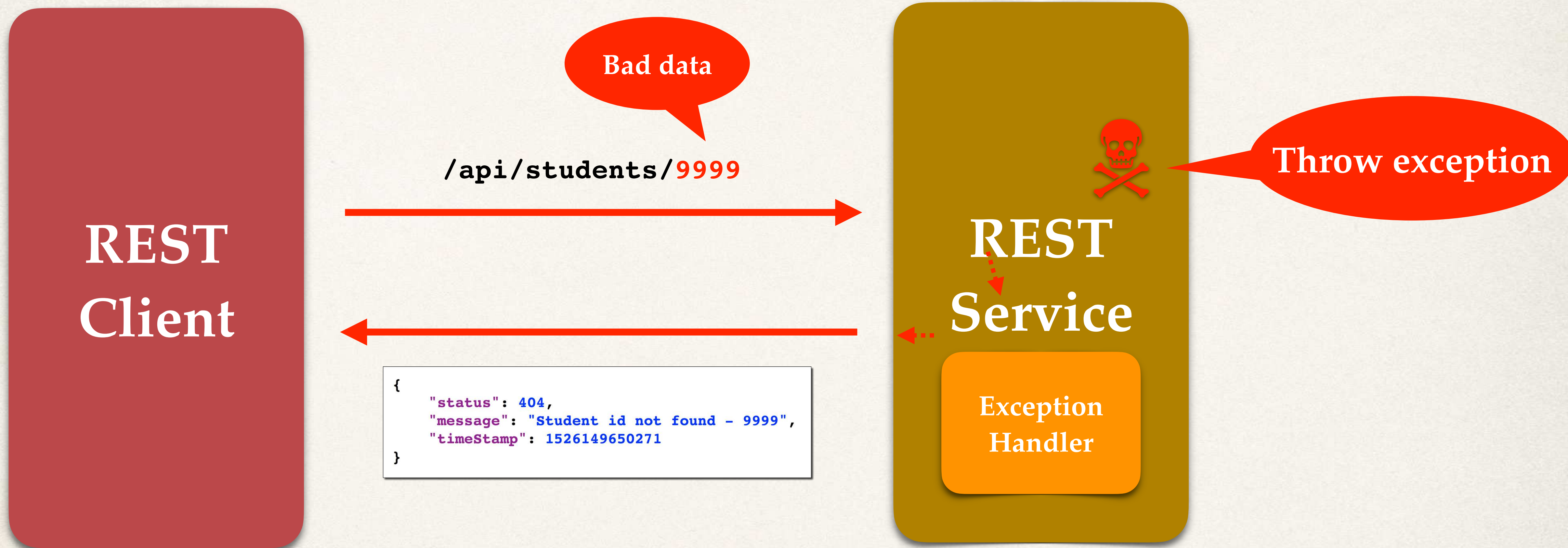


We really want this ...

- Handle the exception and return error as JSON

```
{  
  "status": 404,  
  "message": "Student id not found - 9999",  
  "timeStamp": 1526149650271  
}
```

Spring REST Exception Handling



Development Process

Step-By-Step

1. Create a custom error response class
2. Create a custom exception class
3. Update REST service to throw exception if student not found
4. Add an exception handler method using `@ExceptionHandler`

Step 1: Create custom error response class

- The custom error response class will be sent back to client as JSON

- We will define as Java class (POJO)

StudentErrorResponse
status : int message : String timeStamp : long
getStatus() : int setStatus(...) : void ...

You can define any custom fields that you want to track

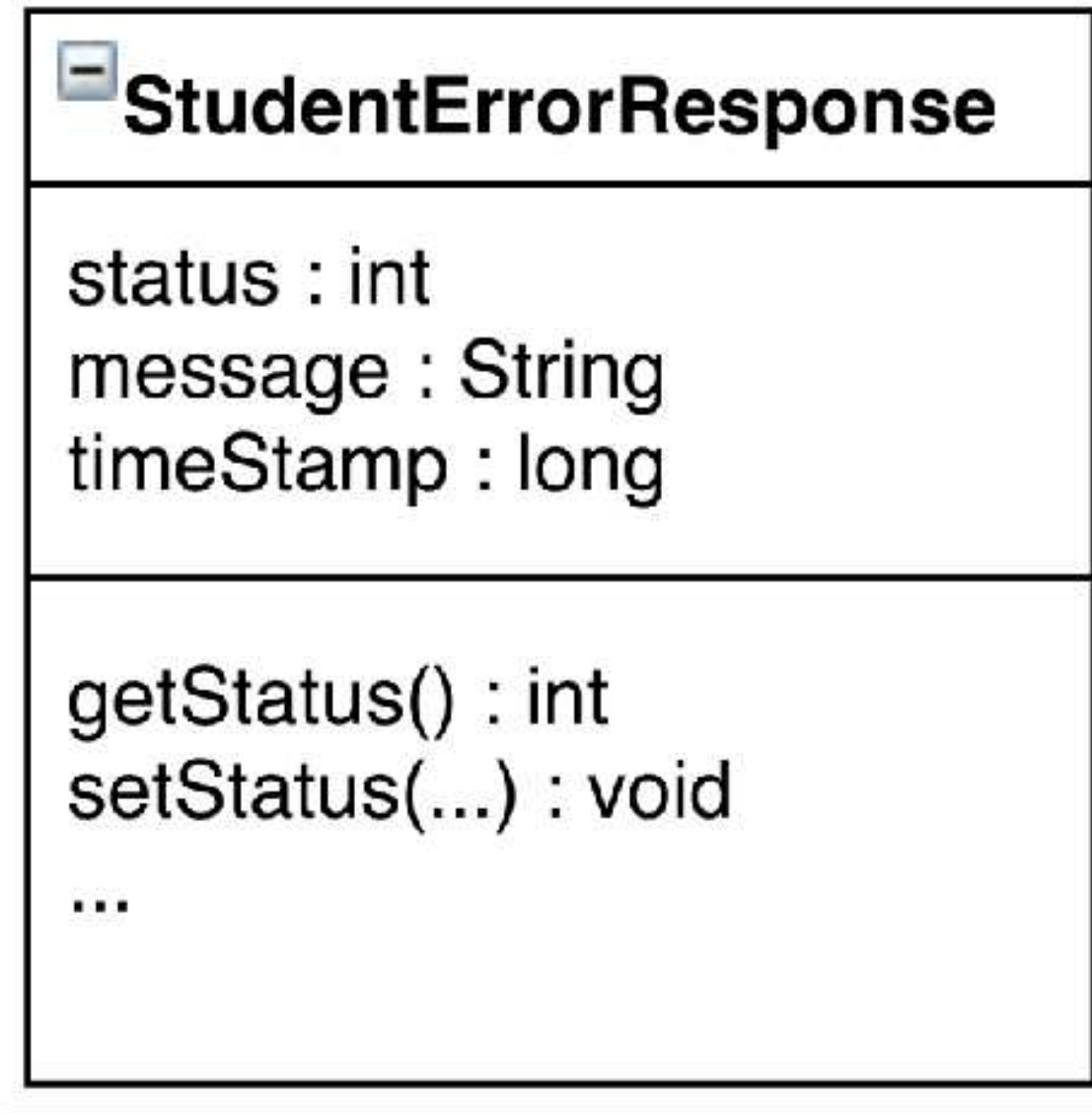
- Jackson will handle converting it to JSON

```
{  
  "status": 404,  
  "message": "Student id not found - 9999",  
  "timeStamp": 1526149650271  
}
```

Step 1: Create custom error response class

File: StudentErrorResponse.java

```
public class StudentErrorResponse {  
  
    private int status;  
    private String message;  
    private long timeStamp;  
  
    // constructors  
  
    // getters / setters  
  
}
```



```
{  
    "status": 404,  
    "message": "Student id not found - 9999",  
    "timeStamp": 1526149650271  
}
```

Step 2: Create custom student exception

- The custom student exception will be used by our REST service
- In our code, if we can't find student, then we'll throw an exception
- Need to define a custom student exception class
 - `StudentNotFoundException`

Step 2: Create custom student exception

File: StudentNotFoundException.java

```
public class StudentNotFoundException extends RuntimeException {  
  
    public StudentNotFoundException(String message) {  
        super(message);  
    }  
  
}
```



Call super class
constructor

Step 3: Update REST service to throw exception

File: StudentRestController.java

```
@RestController
@RequestMapping("/api")
public class StudentRestController {

    @GetMapping("/students/{studentId}")
    public Student getStudent(@PathVariable int studentId) {

        // check the studentId against list size

        if ( (studentId >= theStudents.size()) || (studentId < 0) ) {
            throw new StudentNotFoundException("Student id not found - " + studentId);
        }

        return theStudents.get(studentId);
    }
    ...
}
```

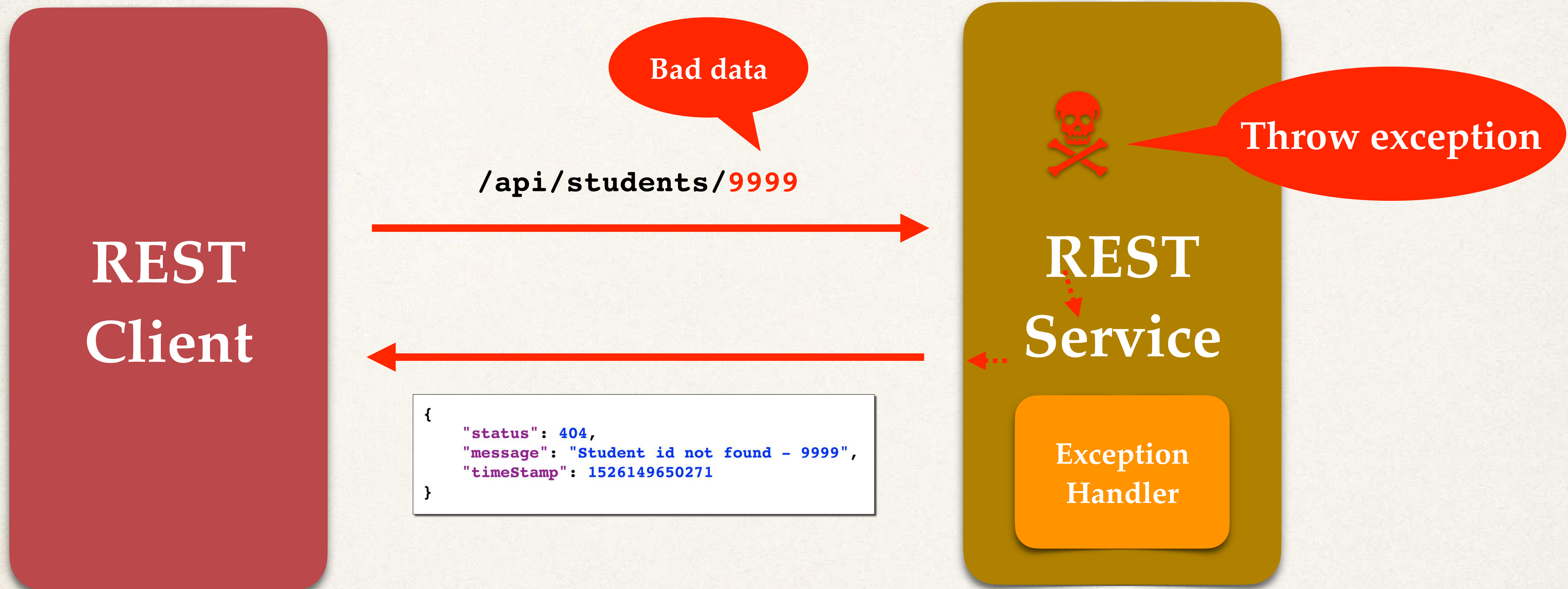
Could also
check results
from DB



Throw exception

Happy path

Spring REST Exception Handling



Step 4: Add exception handler method

- Define exception handler method(s) with `@ExceptionHandler` annotation
- Exception handler will return a `ResponseEntity`
- `ResponseEntity` is a wrapper for the HTTP response object
- `ResponseEntity` provides fine-grained control to specify:
 - HTTP status code, HTTP headers and Response body

Step 4: Add exception handler method

Exception handler method

Type of the response body

Exception type to handle / catch

```
java
@RestController(api)
public class StudentRestController {
    ...

    @ExceptionHandler
    public ResponseEntity<StudentErrorResponse> handleException(StudentNotFoundException exc) {

        StudentErrorResponse error = new StudentErrorResponse();

        error.setStatus(HttpStatus.NOT_FOUND.value());
        error.setMessage(exc.getMessage());
        error.setTimestamp(System.currentTimeMillis());

        return new ResponseEntity<>(error, HttpStatus.NOT_FOUND);
    }
}
```

Body

Status code

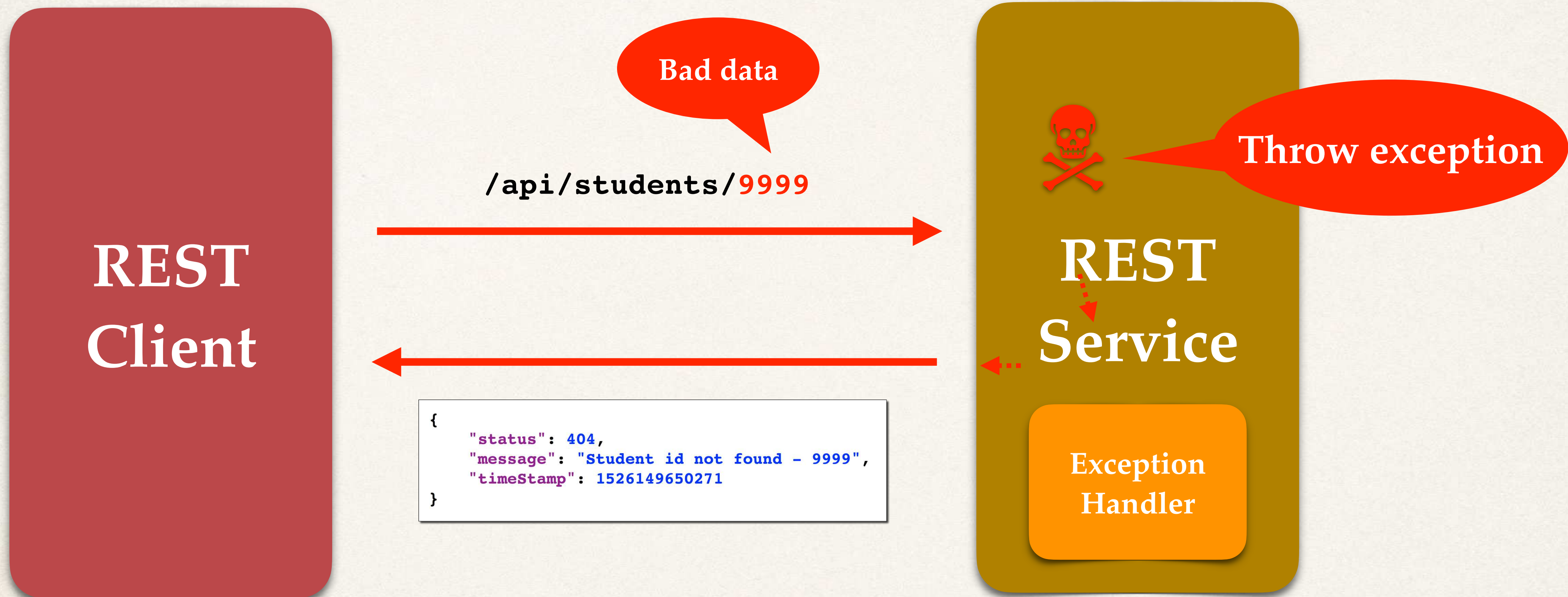
```
{
  "status": 404,
  "message": "Student id not found - 9999",
  "timestamp": 1526149650271
}
```

StudentErrorResponse

status : int
message : String
timestamp : long

getStatus() : int
setStatus(...) : void
...

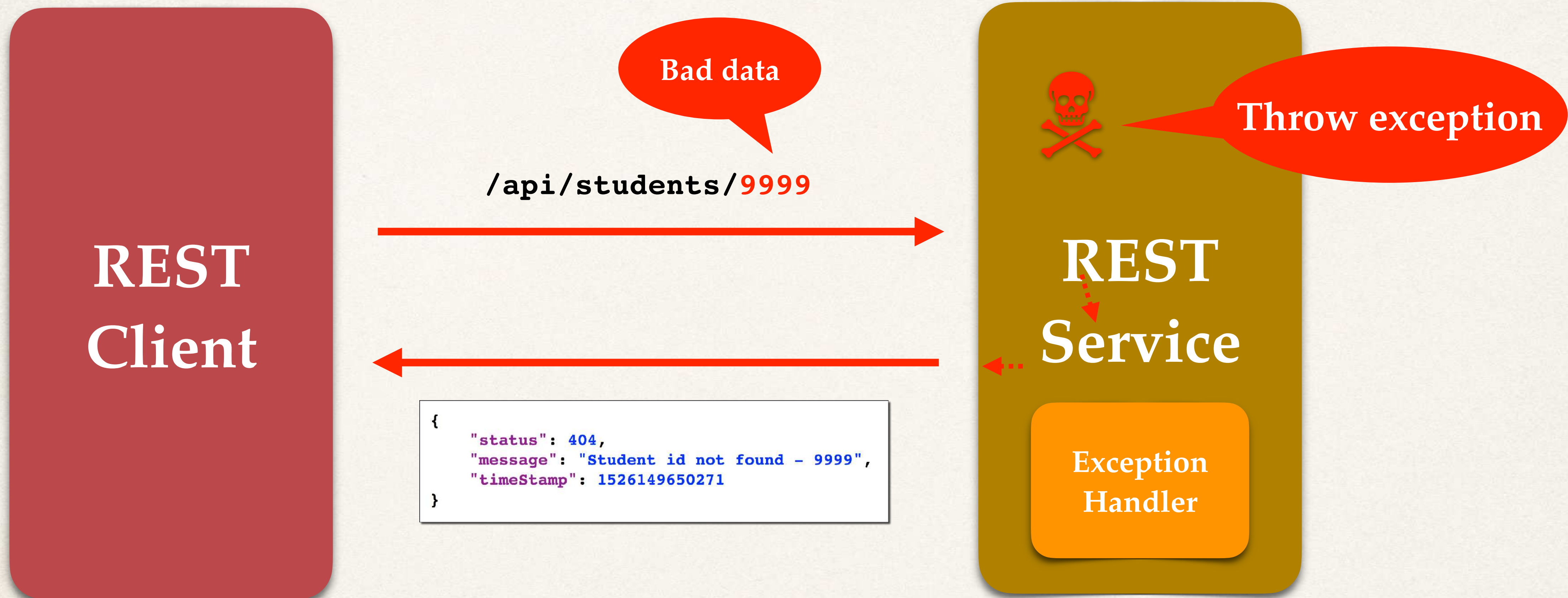
Spring REST Exception Handling



Spring REST - Global Exception Handling

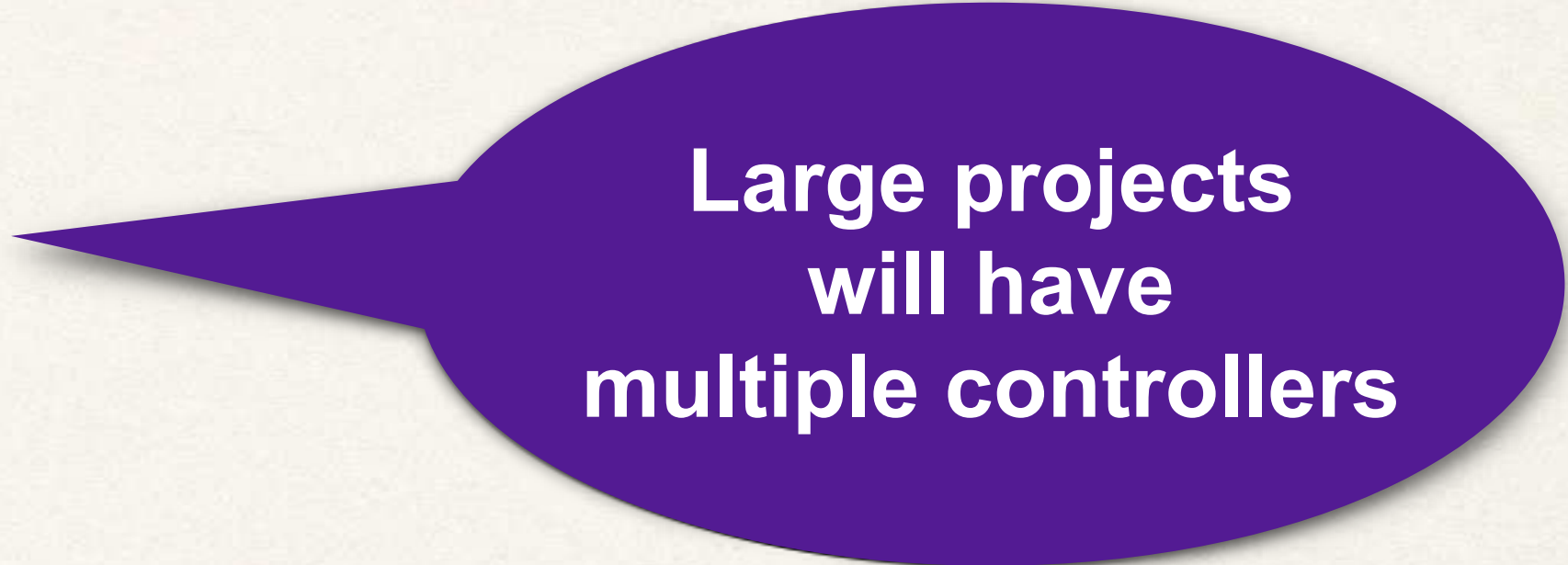


Spring REST Exception Handling



It works, but ...

- Exception handler code is only for the specific REST controller
- Can't be reused by other controllers :-(
- We need **global** exception handlers
 - Promotes reuse
 - Centralizes exception handling



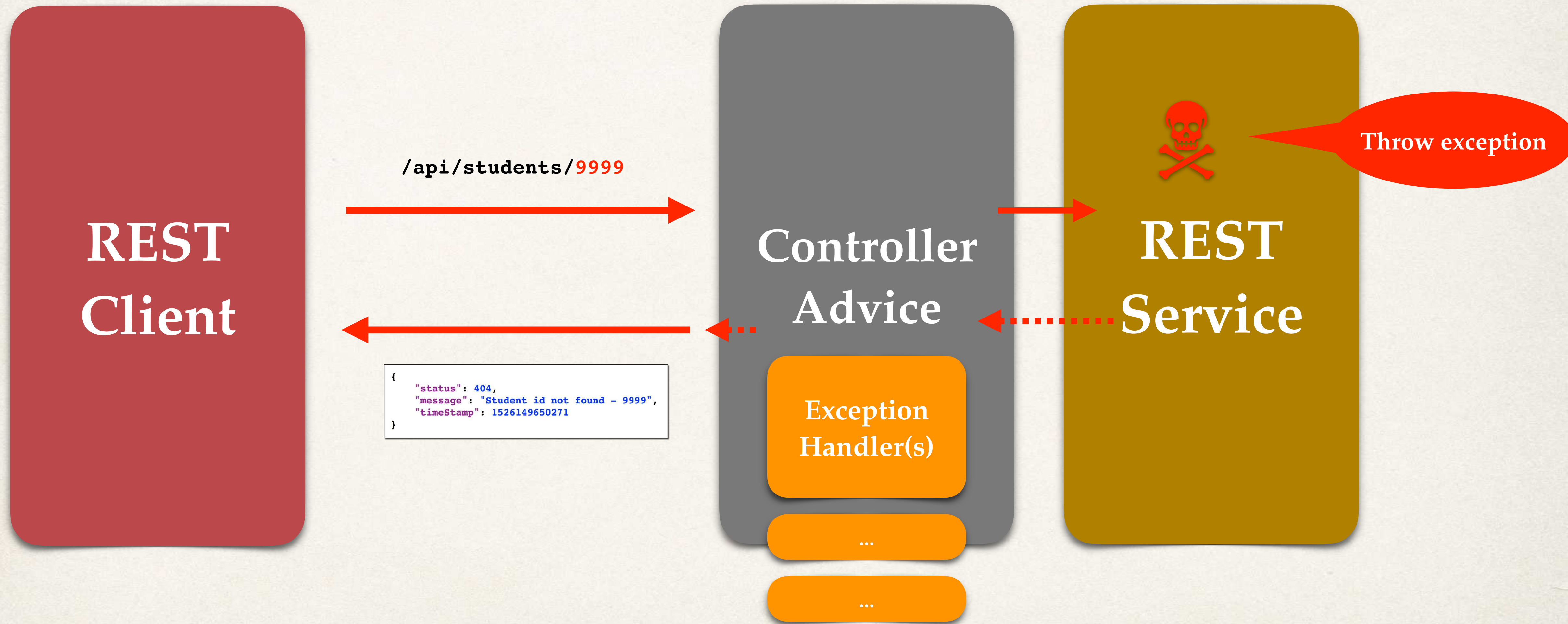
Large projects
will have
multiple controllers

Spring @ControllerAdvice

- @ControllerAdvice is similar to an interceptor / filter
- Pre-process requests to controllers
- Post-process responses to handle exceptions
- Perfect for global exception handling

**Real-time
use of
AOP**

Spring REST Exception Handling



Development Process

Step-By-Step

1. Create new `@ControllerAdvice`
2. Refactor REST service ... remove exception handling code
3. Add exception handling code to `@ControllerAdvice`

Step 1: Create new @ControllerAdvice

File: StudentRestExceptionHandler.java

```
@ControllerAdvice
public class StudentRestExceptionHandler {

    ...

}
```

Step 2: Refactor - remove exception handling

File: StudentRestController.java

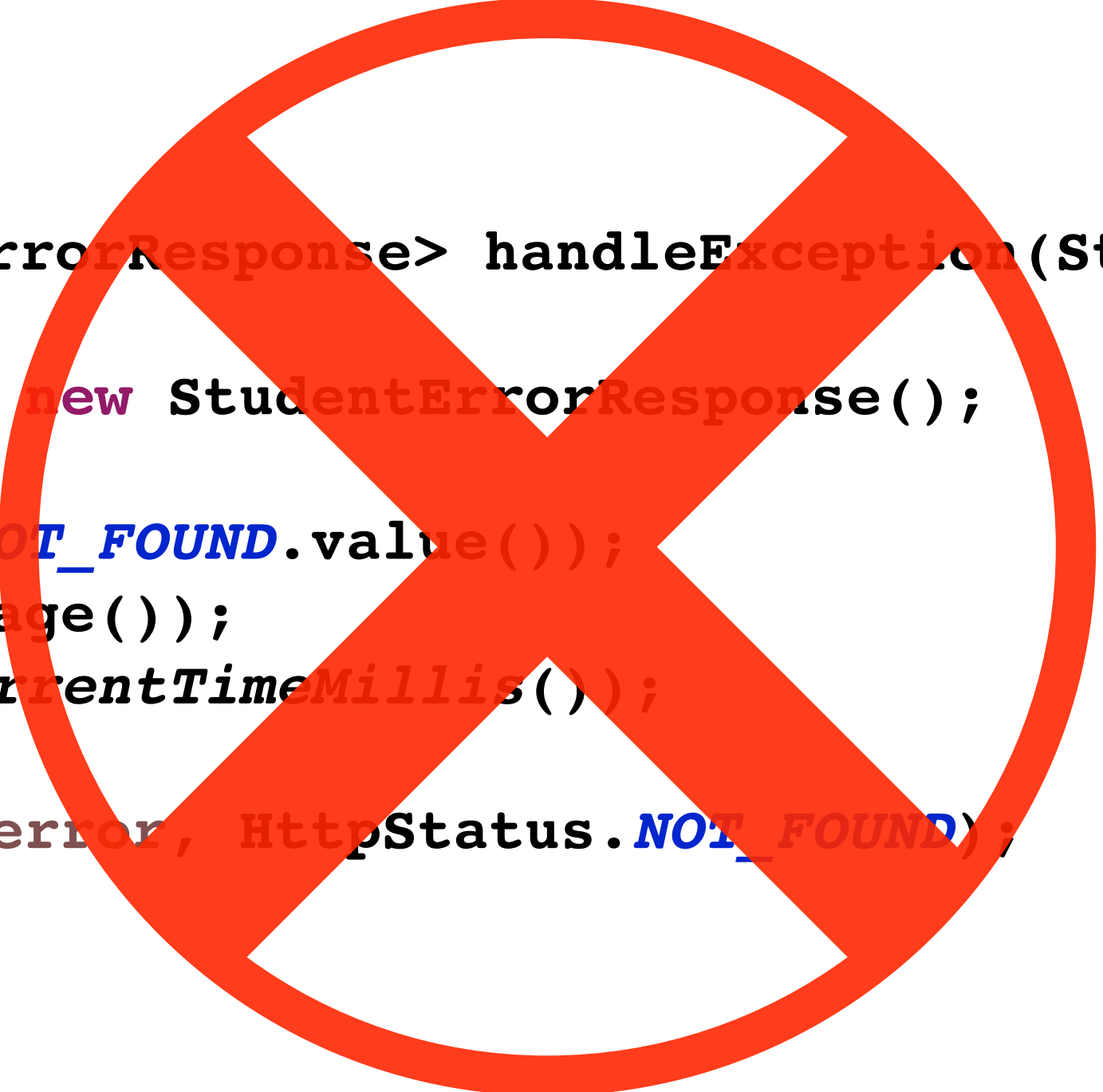
```
@RestController
@RequestMapping("/api")
public class StudentRestController {
    ...

    @ExceptionHandler
    public ResponseEntity<StudentErrorResponse> handleException(StudentNotFoundException exc) {

        StudentErrorResponse error = new StudentErrorResponse();

        error.setStatus(HttpStatus.NOT_FOUND.value());
        error.setMessage(exc.getMessage());
        error.setTimeStamp(System.currentTimeMillis());

        return new ResponseEntity<>(error, HttpStatus.NOT_FOUND);
    }
}
```



Remove
this code

Step 3: Add exception handler to @ControllerAdvice

File: StudentRestExceptionHandler.java

Same code
as before

```
@ControllerAdvice
public class StudentRestExceptionHandler {

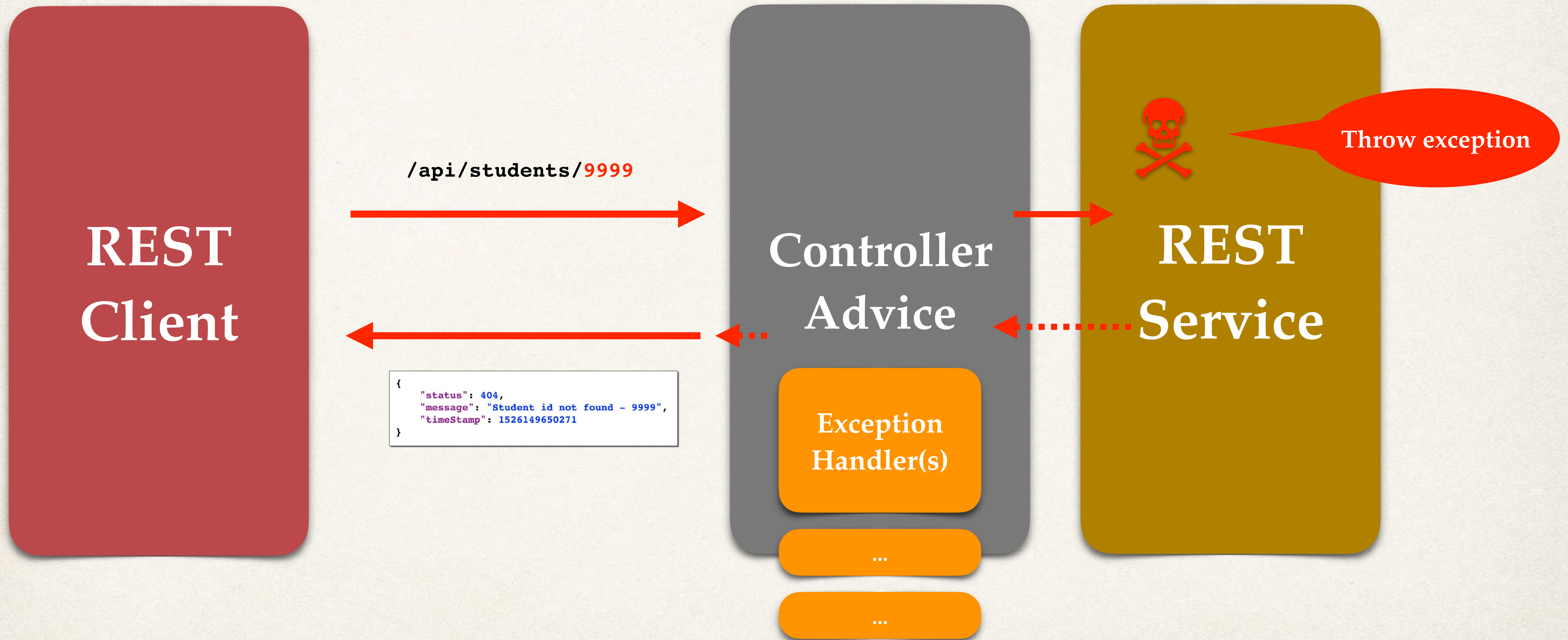
    @ExceptionHandler
    public ResponseEntity<StudentErrorResponse> handleException(StudentNotFoundException exc) {

        StudentErrorResponse error = new StudentErrorResponse();

        error.setStatus(HttpStatus.NOT_FOUND.value());
        error.setMessage(exc.getMessage());
        error.setTimestamp(System.currentTimeMillis());

        return new ResponseEntity<>(error, HttpStatus.NOT_FOUND);
    }
}
```

Spring REST Exception Handling



Spring REST API Design



REST API Design

- For real-time projects, who will use your API?
- Also, how will they use your API?
- Design the API based on requirements

API Design Process

Step-By-Step

1. Review API requirements
2. Identify main resource / entity
3. Use HTTP methods to assign action on resource

API Requirements

From the Boss

Create a REST API for the Employee Directory

REST clients should be able to

- Get a list of employees
- Get a single employee by id
- Add a new employee
- Update an employee
- Delete an employee

Step 2: Identify main resource / entity

- To identify main resource / entity, look for the most prominent "noun"
- For our project, it is "employee"
- Convention is to use plural form of resource / entity: **employees**

`/api/employees`

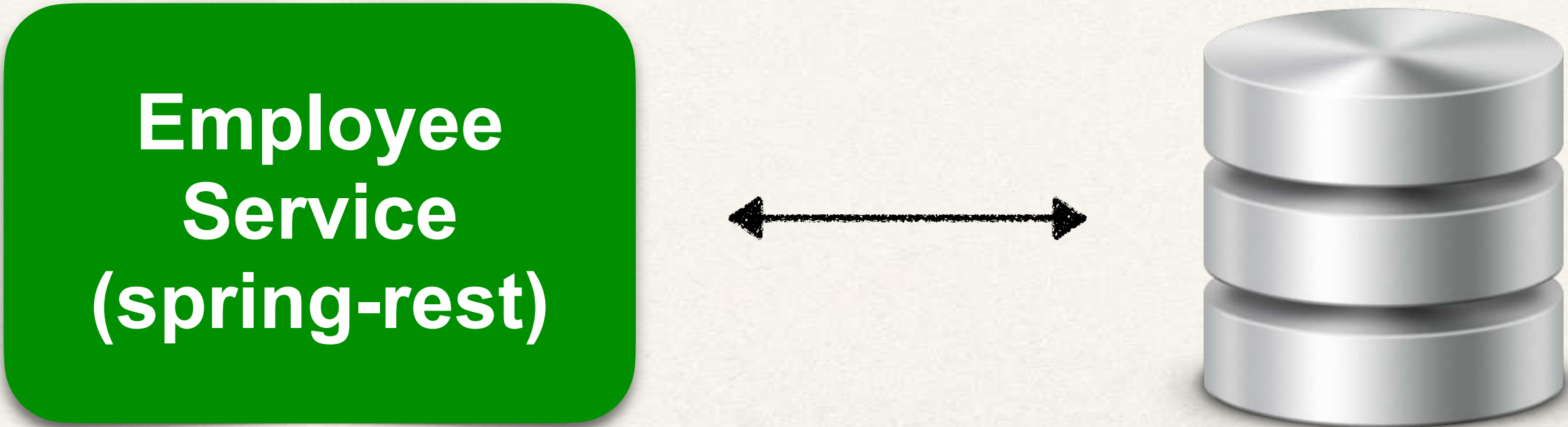
Step 3: Use HTTP methods to assign action on resource

HTTP Method	CRUD Action
POST	<u>C</u> reate a new entity
GET	<u>R</u> ead a list of entities or single entity
PUT	<u>U</u> pdate an existing entity
DELETE	<u>D</u> eleate an existing entity

Full
CRUD

Employee Real-Time Project

HTTP Method	Endpoint	CRUD Action
POST	/api/employees	<u>C</u> reate a new employee
GET	/api/employees	<u>R</u> ead a list of employees
GET	/api/employees/{employeeId}	<u>R</u> ead a single employee
PUT	/api/employees	<u>U</u> pdate an existing employee
DELETE	/api/employees/{employeeId}	<u>D</u> delete an existing employee



Anti-Patterns

- DO NOT DO THIS ... these are REST anti-patterns, bad practice



`/api/employeesList`
`/api/deleteEmployee`
`/api/addEmployee`
`/api/updateEmployee`

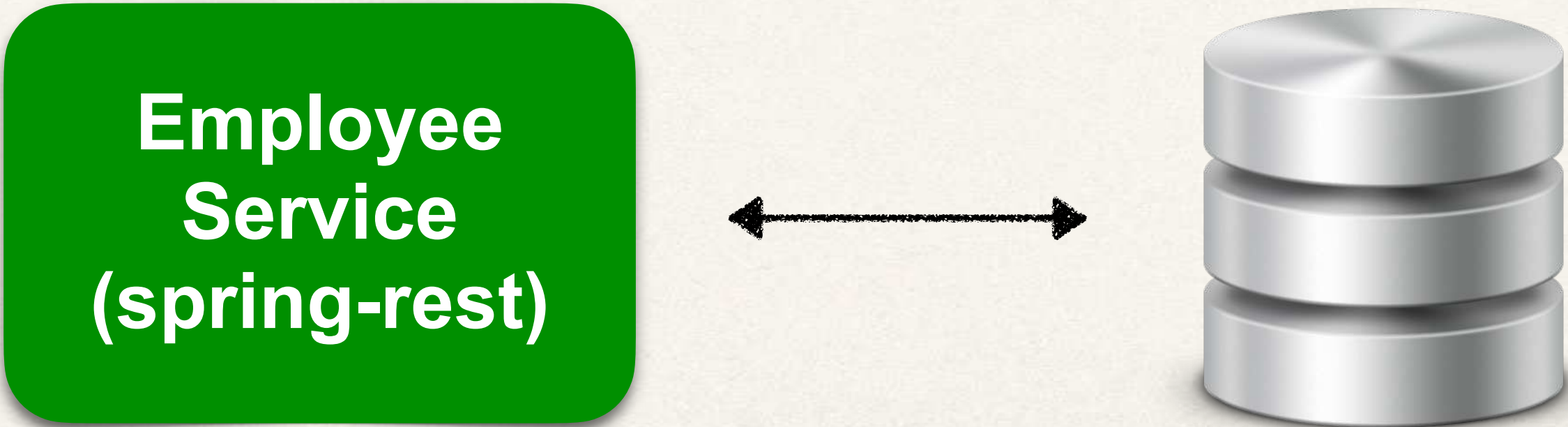
Don't include actions in the endpoint



Instead, use
HTTP methods
to assign actions

Employee Real-Time Project

HTTP Method	Endpoint	CRUD Action
POST	/api/employees	<u>C</u> reate a new employee
GET	/api/employees	<u>R</u> ead a list of employees
GET	/api/employees/{employeeId}	<u>R</u> ead a single employee
PUT	/api/employees	<u>U</u> pdate an existing employee
DELETE	/api/employees/{employeeId}	<u>D</u> delete an existing employee



More API Examples

- On the following slides, we'll look at APIs from other real-time projects
- PayPal
- GitHub
- Salesforce

PayPal



- PayPal Invoicing API
 - <https://developer.paypal.com/docs/api/invoicing/>

PayPal Developer

Docs

APIs

Support

Create draft invoice

POST

/v1/invoicing/invoices

List invoices

GET

/v1/invoicing/invoices

Show invoice details

GET

/v1/invoicing/invoices/{invoice_id}

Update invoice

PUT

/v1/invoicing/invoices/{invoice_id}

Delete draft invoice

DELETE

/v1/invoicing/invoices/{invoice_id}

GitHub



- GitHub Repositories API
 - `https://developer.github.com/v3/repos/#repositories`

GitHub Developer

Create a new repository

POST /user/repos

Delete a repository

DELETE /repos/:owner/:repo

List your repositories

GET /user/repos

Get a repository

GET /repos/:owner/:repo

SalesForce REST API



- Industries REST API
- <https://sforce.co/2J40ALH>

Retrieve All Individuals

GET /services/apexrest/v1/individual/

Retrieve One Individual

GET /services/apexrest/v1/individual/{individual_id}

Create an individual

POST /services/apexrest/clinic01/v1/individual/

Update an individual

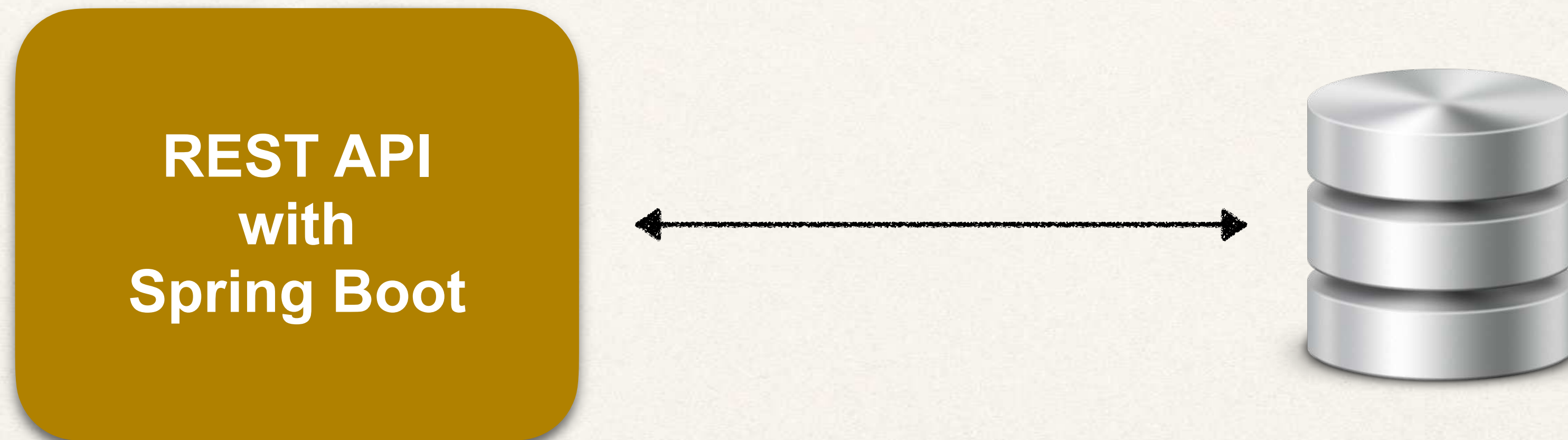
PUT /services/apexrest/clinic01/v1/individual/

Spring Boot REST API - Real Time Project



Real-Time Project

REST API with Spring Boot that connects to a database



API Requirements

From the Boss

Create a REST API for the Employee Directory

REST clients should be able to

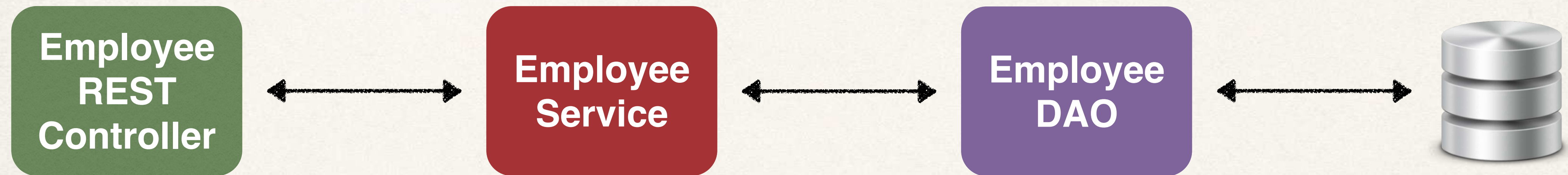
- Get a list of employees
- Get a single employee by id
- Add a new employee
- Update an employee
- Delete an employee

Development Process

Step-By-Step

1. Set up Database Dev Environment
2. Create Spring Boot project using Spring Initializr
3. Get list of employees
4. Get single employee by ID
5. Add a new employee
6. Update an existing employee
7. Delete an existing employee

Application Architecture



Setup Database Table



employee.sql

1. File: employee.sql
2. Create a new database table: **employee**
3. Load table with sample data

Details on downloading the SQL file
provided in next section



Creating Spring Boot Project



Development Process

Step-By-Step

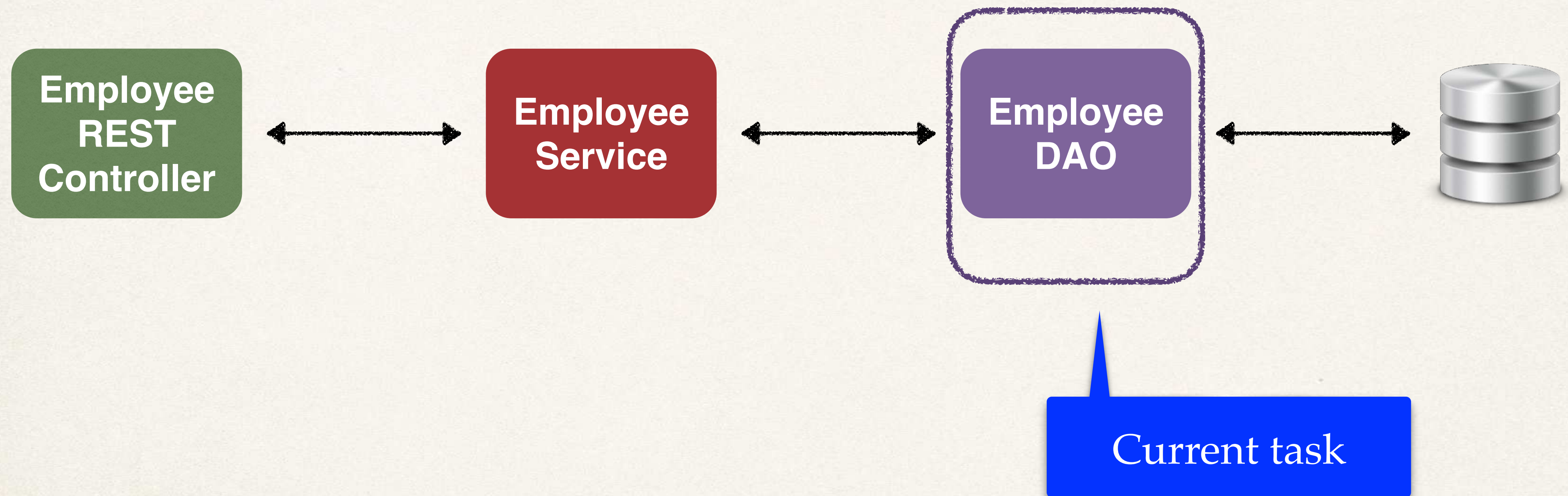
1. Set up Database Dev Environment
2. Create Spring Boot project using Spring Initializr
3. Get list of employees
4. Get single employee by ID
5. Add a new employee
6. Update an existing employee
7. Delete an existing employee

Our Task

Create DAO in Spring Boot



Application Architecture



Development Process

1. Set up Database Dev Environment
2. Create Spring Boot project using Spring Initializr

3. Get list of employees
4. Get single employee by ID
5. Add a new employee
6. Update an existing employee
7. Delete an existing employee

Step-By-Step

**Let's build a
DAO layer for this**

DAO Interface

```
public interface EmployeeDAO {  
  
    List<Employee> findAll();  
  
}
```

DAO Impl

Same interface for
consistent API

```
@Repository
public class EmployeeDAOJpaImpl implements EmployeeDAO {

    private EntityManager entityManager;

    @Autowired
    public EmployeeDAOJpaImpl(EntityManager theEntityManager) {
        entityManager = theEntityManager;
    }

    ...
}
```

Automatically created
by Spring Boot

Constructor
injection

Get a list of employees

```
@Override
public List<Employee> findAll() {

    // create a query
    TypedQuery<Employee> theQuery =
        entityManager.createQuery("from Employee", Employee.class);

    // execute query and get result list
    List<Employee> employees = theQuery.getResultList();

    // return the results
    return employees;
}
```

Development Process

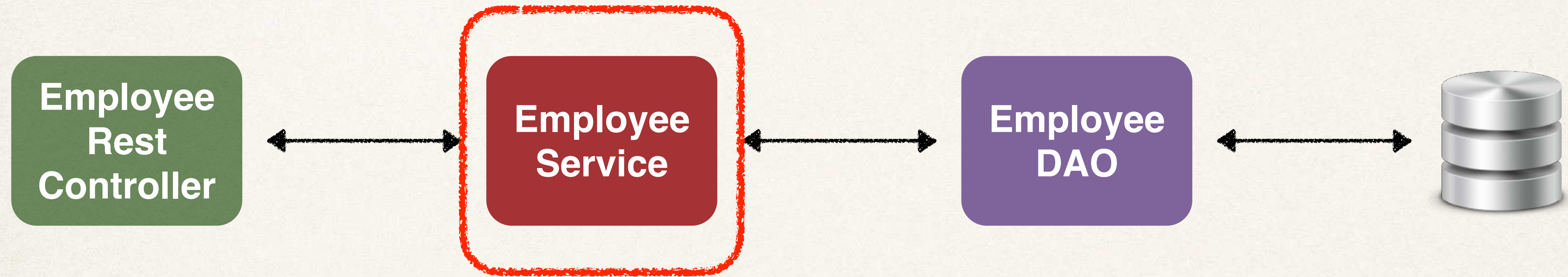
Step-By-Step

1. Update db configs in application.properties
2. Create Employee entity
3. Create DAO interface
4. Create DAO implementation
5. Create REST controller to use DAO

Define Services with @Service

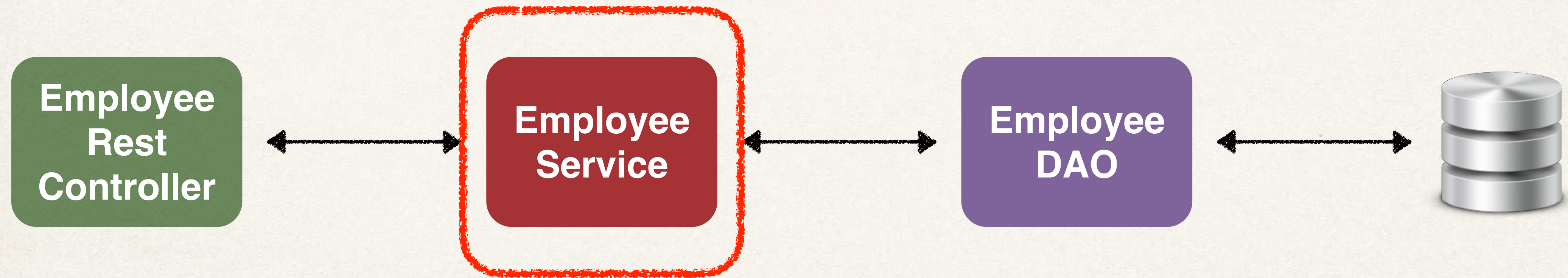


Refactor: Add a Service Layer

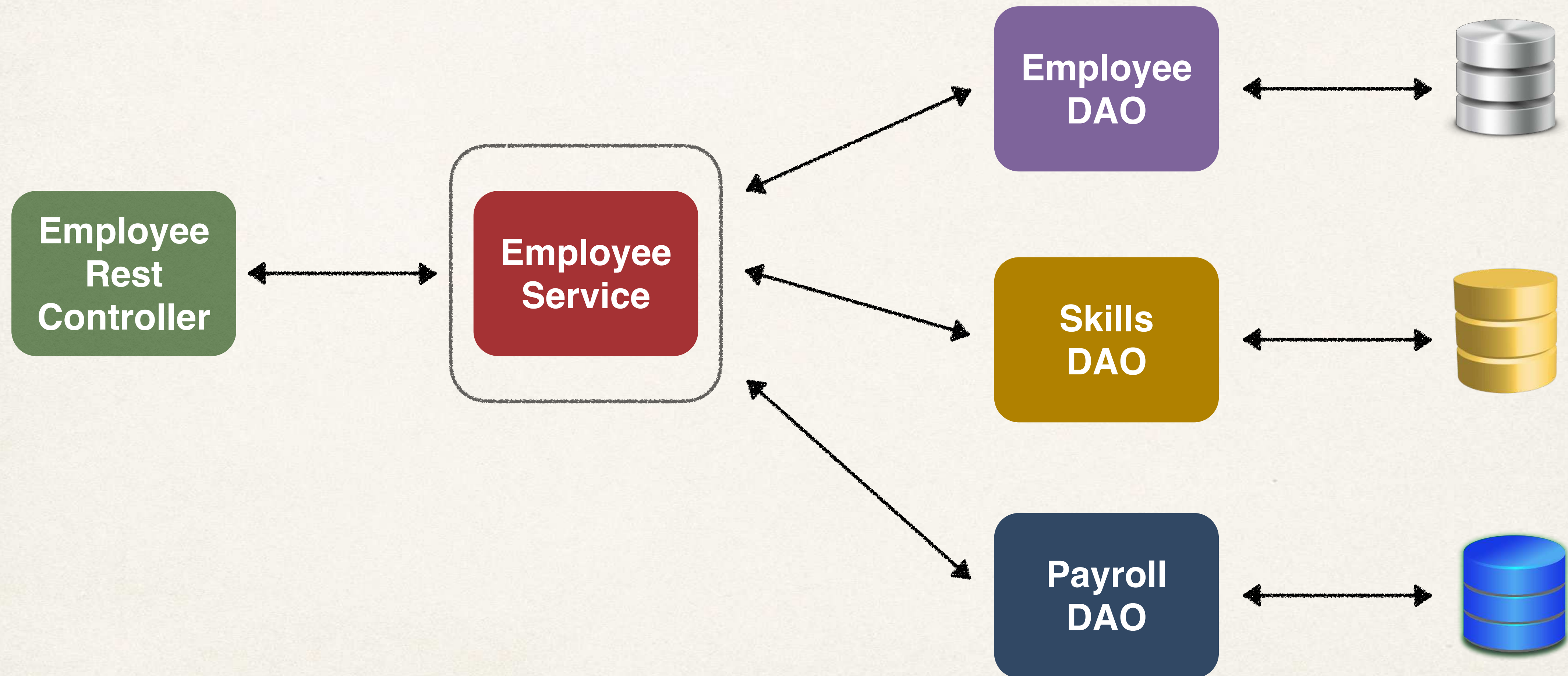


Purpose of Service Layer

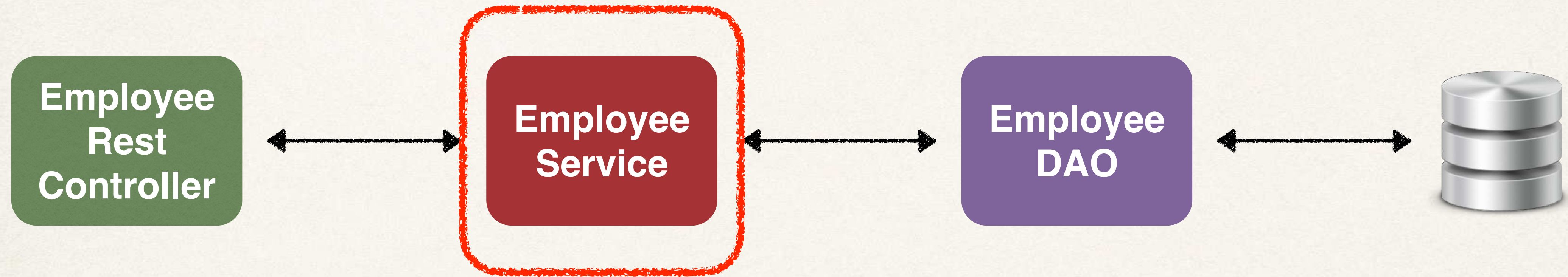
- ❖ Service Facade design pattern
- ❖ Intermediate layer for custom business logic
- ❖ Integrate data from multiple sources (DAO/repositories)



Integrate Multiple Data Sources

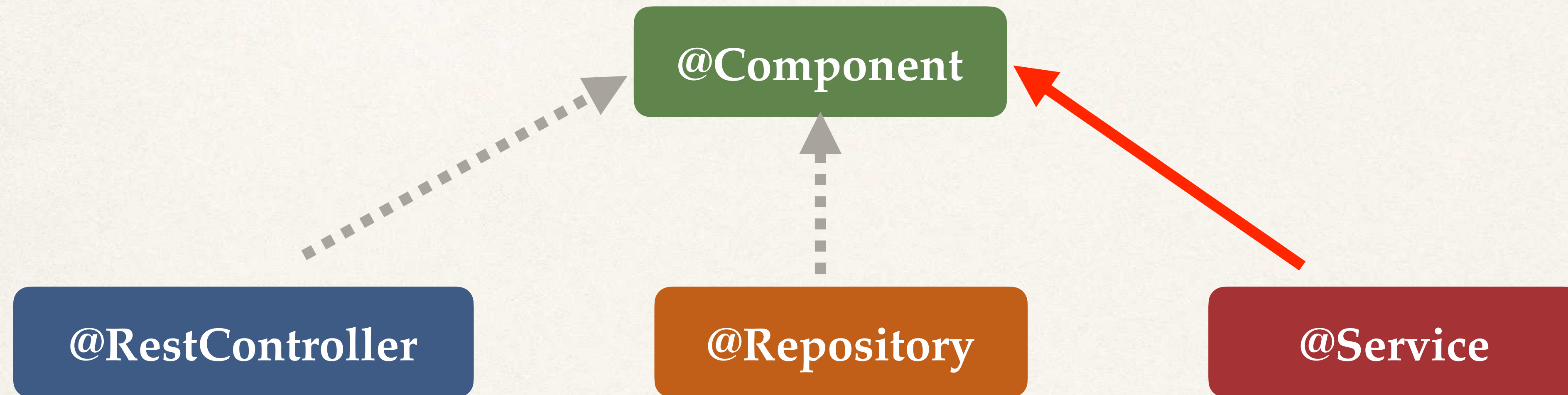


Most Times - Delegate Calls



Specialized Annotation for Services

- Spring provides the **@Service** annotation



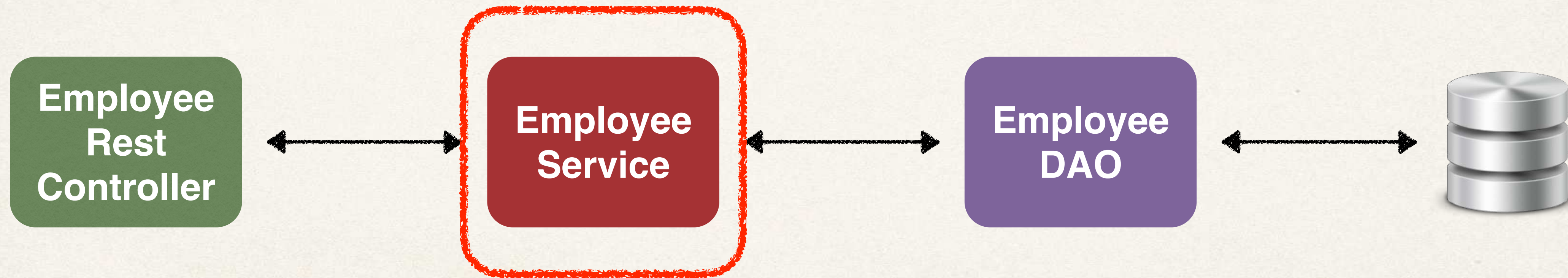
Specialized Annotation for Services

- **@Service** applied to Service implementations
- Spring will automatically register the Service implementation
 - thanks to component-scanning

Employee Service

1. Define Service interface
2. Define Service implementation
 - Inject the EmployeeDAO

Step-By-Step



Step 1: Define Service interface

```
public interface EmployeeService {  
  
    List<Employee> findAll();  
  
}
```

Step 2: Define Service implementation

@Service - enables component scanning

@Service

```
public class EmployeeServiceImpl implements EmployeeService {
```

```
// inject EmployeeDAO ...
```

@Override

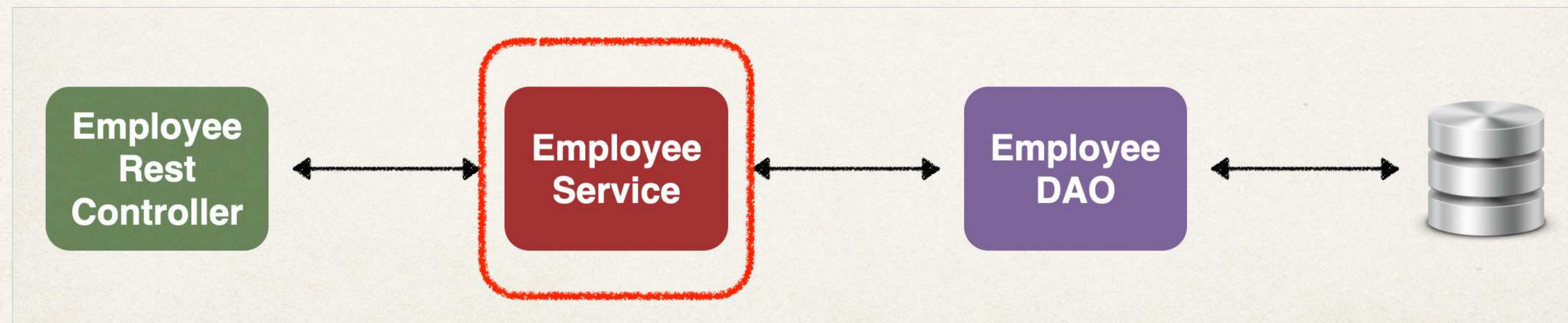
```
public List<Employee> findAll() {  
    return employeeDAO.findAll();  
}  
}
```

DAO: Find, Add, Update and Delete



Service Layer - Best Practice

- ❖ Best practice is to apply transactional boundaries at the service layer
- ❖ It is the service layer's responsibility to manage transaction boundaries
- ❖ For implementation code
 - ❖ Apply @Transactional on service methods
 - ❖ Remove @Transactional on DAO methods if they already exist

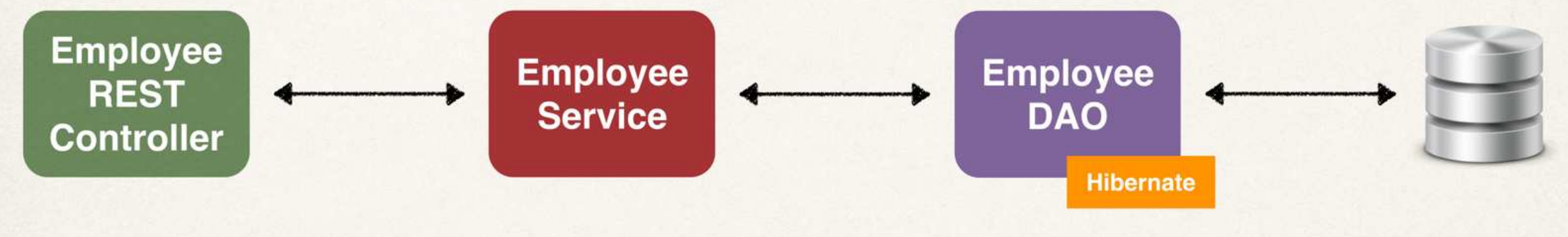


Development Process

Step-By-Step

1. Set up Database Dev Environment
2. Create Spring Boot project using Spring Initializr
3. Get list of employees
4. Get single employee by ID
5. Add a new employee
6. Update an existing employee
7. Delete an existing employee

DAO methods



DAO: Get a single employee

```
@Override
public Employee findById(int theId) {

    // get employee
    Employee theEmployee = entityManager.find(Employee.class, theId);

    // return employee
    return theEmployee;
}
```



DAO: Add or Update employee

Note: We don't use @Transactional at DAO layer
It will be handled at Service layer

```
@Override
public Employee save(Employee theEmployee) {

    // save or update the employee
    Employee dbEmployee = entityManager.merge(theEmployee);

    // return dbEmployee
    return dbEmployee;
}
```

if id == 0
then save/insert
else update

Return dbEmployee
It has updated id from the database
(in the case of insert)

DAO: Delete an existing employee

Note: We don't use @Transactional at DAO layer
It will be handled at Service layer

```
@Override
public void deleteById(int theId) {

    // find the employee by id
    Employee theEmployee = entityManager.find(Employee.class, theId);

    // delete the employee
    entityManager.remove(theEmployee);
}
```



DAO: Find, Add, Update and Delete

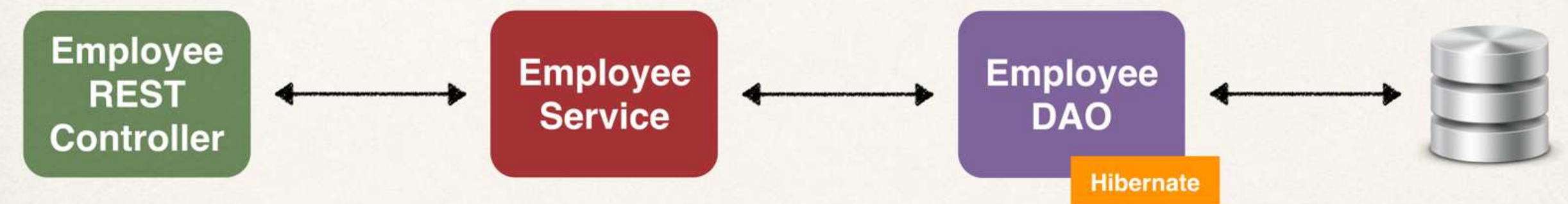


Development Process

Step-By-Step

1. Set up Database Dev Environment
2. Create Spring Boot project using Spring Initializr
3. Get list of employees
4. Get single employee by ID
5. Add a new employee
6. Update an existing employee
7. Delete an existing employee

DAO methods



Service: Find, Add, Update and Delete

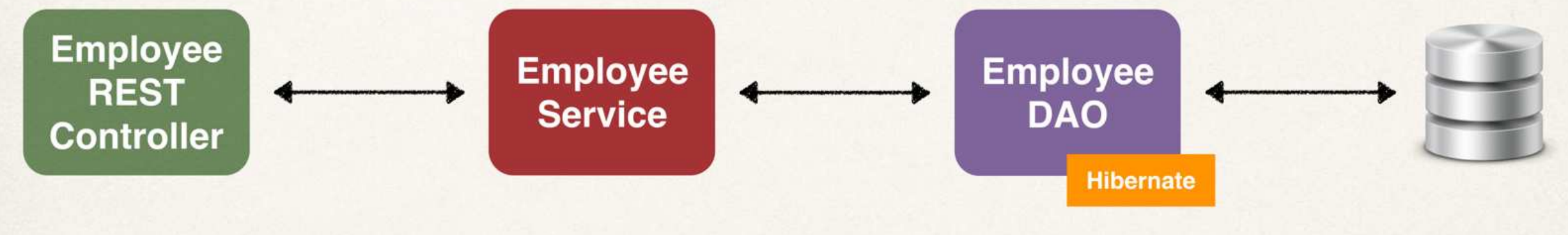


Development Process

Step-By-Step

1. Set up Database Dev Environment
2. Create Spring Boot project using Spring Initializr
3. Get list of employees
4. Get single employee by ID
5. Add a new employee
6. Update an existing employee
7. Delete an existing employee

Service methods



Rest Controller Methods - Find and Add

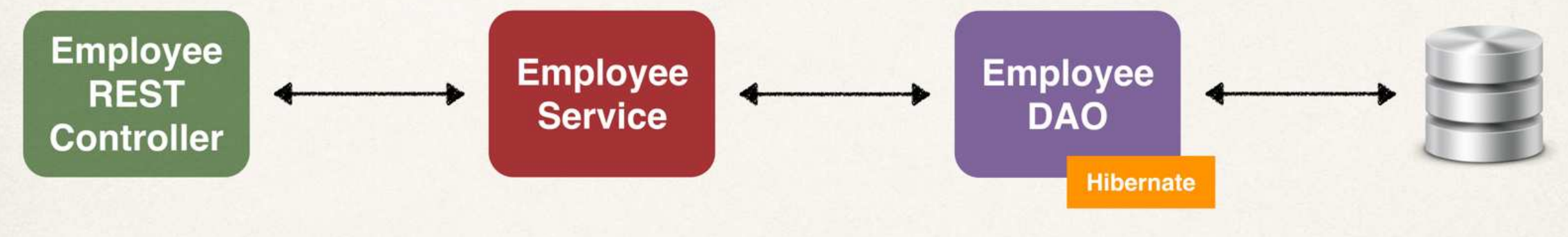


Development Process

Step-By-Step

1. Set up Database Dev Environment
2. Create Spring Boot project using Spring Initializr
3. Get list of employees
4. Get single employee by ID
5. Add a new employee
6. Update an existing employee
7. Delete an existing employee

Rest Controller
methods



Read a Single Employee



Create a New Employee

Since new employee,
we are not
passing id / primary key

REST
Client

POST

/api/employees

```
{  
  "firstName": "Juan",  
  "lastName": "Perez",  
  "email": "juan.perez@luv2code.com"  
}
```

Employee
REST
Controller

Response contains
new id / primary key

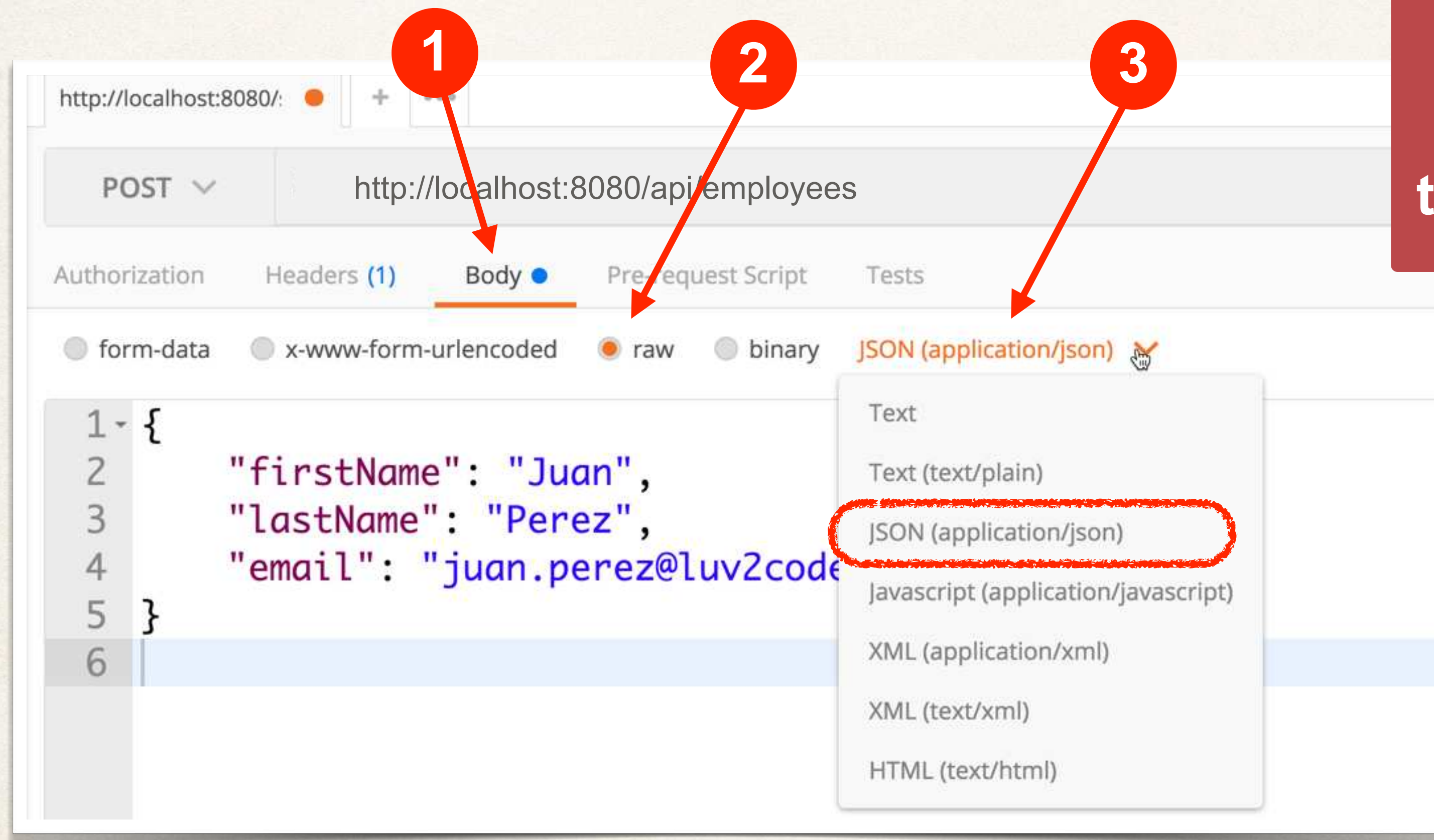
```
{  
  "id": 7,  
  "firstName": "Juan",  
  "lastName": "Perez",  
  "email": "juan.perez@luv2code.com"  
}
```

Sending JSON to Spring REST Controllers

- When sending JSON data to Spring REST Controllers
- For controller to process JSON data, need to set a HTTP request header
 - `Content-type: application/json`
- Need to configure REST client to send the correct HTTP request header

Postman - Sending JSON in Request Body

- Must set HTTP request header in Postman



Based on these configs,
Postman will automatically set
the correct HTTP request header

Rest Controller Methods - Update and Delete

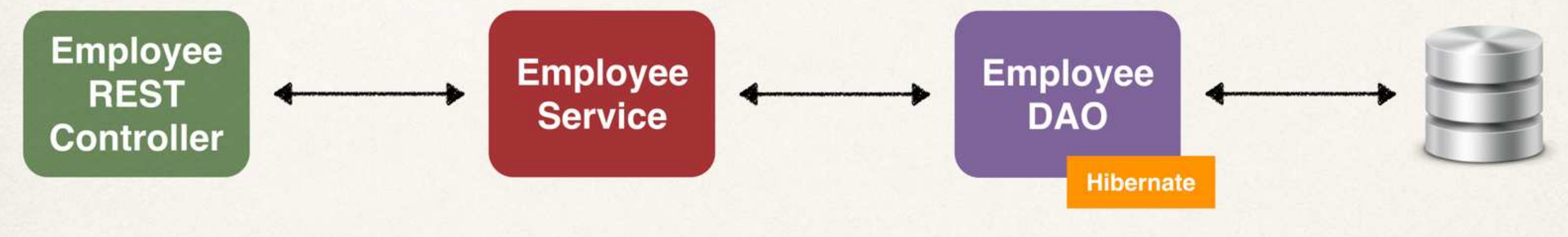


Development Process

Step-By-Step

1. Set up Database Dev Environment
2. Create Spring Boot project using Spring Initializr
3. Get list of employees
4. Get single employee by ID
5. Add a new employee
6. Update an existing employee
7. Delete an existing employee

Rest Controller methods



Update Employee

ID of employee to update
With updated info

PUT

/api/employees

```
{  
  "id" : 1,  
  "firstName" : "Daniel",  
  "lastName" : "Vega",  
  "email" : "daniel.vega@luv2code.com"  
}
```

REST
Client

Employee
REST
Controller

```
{  
  "id" : 1,  
  "firstName" : "Daniel",  
  "lastName" : "Vega",  
  "email" : "daniel.vega@luv2code.com"  
}
```

Response contains
updated info (echoed)

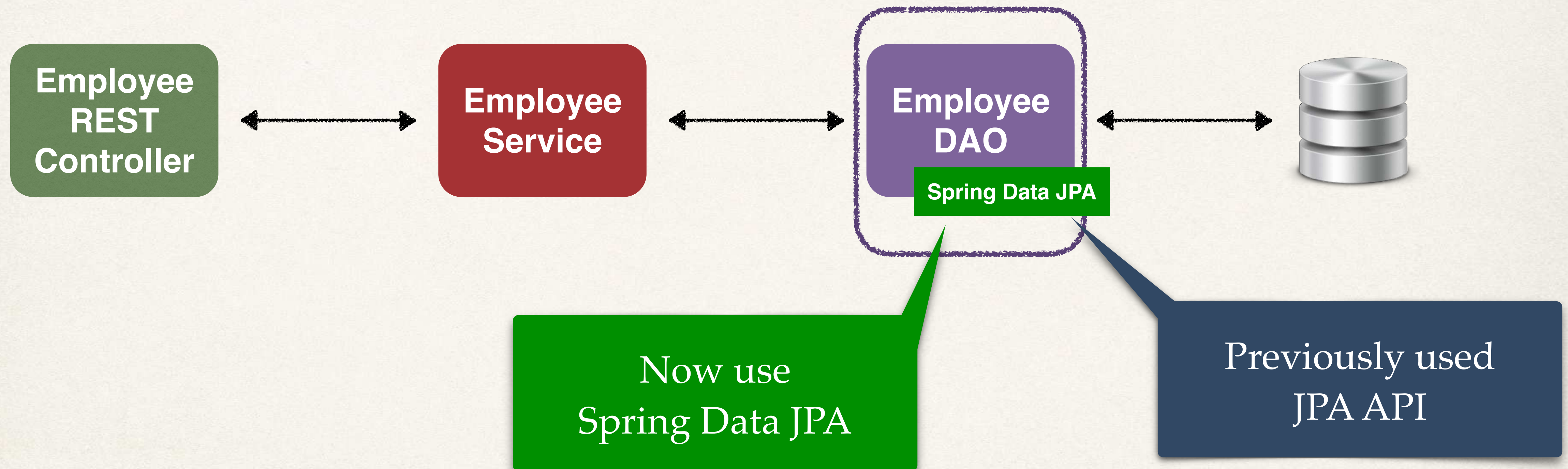
Delete Employee



Spring Data JPA in Spring Boot



Application Architecture



The Problem

- We saw how to create a DAO for **Employee**
- What if we need to create a DAO for another entity?
- **Customer, Student, Product, Book ...**
- Do we have to repeat all of the same code again???

```
public interface EmployeeDAO {  
  
    public List<Employee> findAll();  
  
    public Employee findById(int theId);  
  
    public void save(Employee theEmployee);  
  
    public void deleteById(int theId);  
  
}
```

```
@Repository  
public class EmployeeDAOJpaImpl implements EmployeeDAO {  
  
    private EntityManager entityManager;  
  
    @Autowired  
    public EmployeeDAOJpaImpl(EntityManager theEntityManager) {  
        entityManager = theEntityManager;  
    }  
  
    @Override  
    public List<Employee> findAll() {  
  
        // create a query  
        TypedQuery<Employee> theQuery =  
            entityManager.createQuery("from Employee", Employee.class);  
  
        // execute query and get result list  
        List<Employee> employees = theQuery.getResultList();  
  
        // return the results  
        return employees;  
    }  
  
    @Override  
    public Employee findById(int theId) {  
  
        // get employee  
        Employee theEmployee =  
            entityManager.find(Employee.class, theId);  
  
        // return employee  
        return theEmployee;  
    }  
}
```

Creating DAO

- You may have noticed a pattern with creating DAOs

```
@Override
public Employee findById(int theId) {

    // get data
    Employee theData = entityManager.find(Employee.class, theId);

    // return data
    return theData;
}
```

Most of the code
is the same

Only difference is the
entity type and primary key

Entity type

Primary key

My Wish

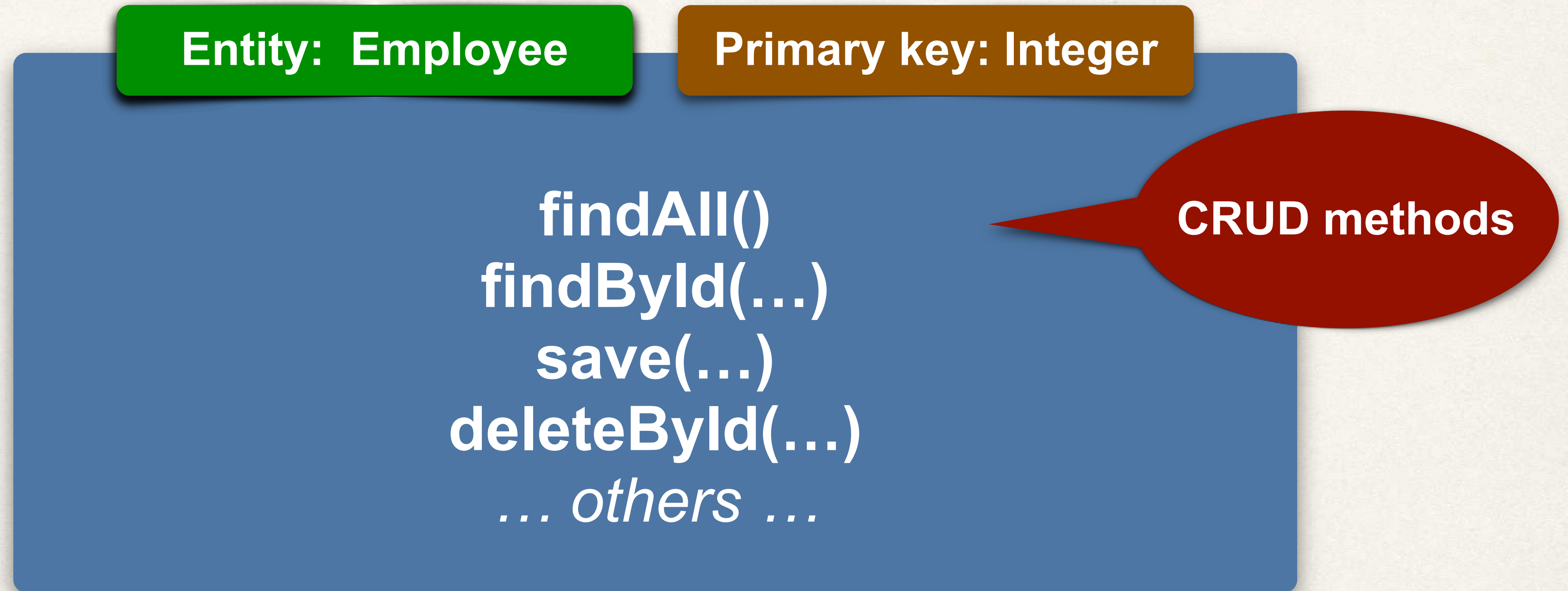
- I wish we could tell Spring:

Create a DAO for me

Plug in my entity type and primary key

Give me all of the basic CRUD features for free

My Wish Diagram



Spring Data JPA - Solution

- Spring Data JPA is the solution!!!! <https://spring.io/projects/spring-data-jpa>
- Create a DAO and just plug in your entity type and primary key
- Spring will give you a CRUD implementation for FREE like MAGIC!!
- Helps to minimize boiler-plate DAO code ... yaaay!!!

More than 70% reduction in code ... depending on use case



JpaRepository

- Spring Data JPA provides the interface: **JpaRepository**
- Exposes methods (some by inheritance from parents)

Entity: Employee

Primary key: Integer

findAll()
findById(...)
save(...)
deleteById(...)
... others ...



Development Process

Step-By-Step

1. Extend **JpaRepository** interface
2. Use your Repository in your app

No need for
implementation class

Step 1: Extend JpaRepository interface

Entity type

Primary key

```
public interface EmployeeRepository extends JpaRepository<Employee, Integer> {  
  
    // that's it ... no need to write any code LOL!  
  
}
```

No need for
implementation class

Get these
methods for free



Entity: Employee

Primary key: Integer

findAll()
findById(...)
save(...)
deleteById(...)
... others ...

JpaRepository Docs

- Full list of methods available ... see JavaDoc for **JpaRepository**

www.luv2code.com/jpa-repository-javadoc

Step 2: Use Repository in your app

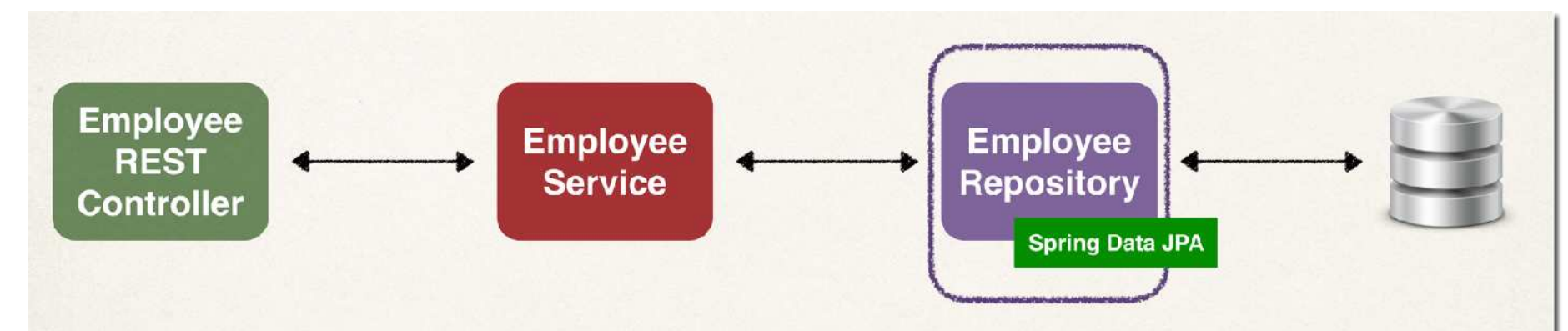
```
@Service
public class EmployeeServiceImpl implements EmployeeService {

    private EmployeeRepository employeeRepository;

    @Autowired
    public EmployeeServiceImpl(EmployeeRepository theEmployeeRepository) {
        employeeRepository = theEmployeeRepository;
    }

    @Override
    public List<Employee> findAll() {
        return employeeRepository.findAll();
    }
    ...
}
```

Our repository



Magic method that is available via repository



Minimized Boilerplate Code

Before Spring Data JPA

```
public interface EmployeeDAO {  
    public List<Employee> findAll();  
    public Employee findById(int theId);  
    public void save(Employee theEmployee);  
    public void delete(int theId);  
}
```

```
@Repository  
public class EmployeeDAOJpaImpl implements EmployeeDAO {  
    private EntityManager em;  
    @Autowired  
    public EmployeeDAOJpaImpl(EntityManager theEntityManager) {  
        em = theEntityManager;  
    }  
    @Override  
    public List<Employee> findAll() {  
        // create TypedQuery  
        TypedQuery<Employee> query = em.createQuery("select e from Employee e", Employee.class);  
        // execute query  
        List<Employee> results = query.getResultList();  
        // return results  
        return results;  
    }  
    @Override  
    public Employee findById(int theId) {  
        // get employee  
        Employee theEmployee = em.find(Employee.class, theId);  
        // return employee  
        return theEmployee;  
    }  
}
```

**2 Files
30+ lines of code**

After Spring Data JPA

```
public interface EmployeeRepository extends JpaRepository<Employee, Integer> {  
    // that's it ... no need to write any code LOL!  
}
```

**1 File
3 lines of code!**

**No need for
implementation class**



Advanced Features

- Advanced features available for
 - Extending and adding custom queries with JPQL
 - Query Domain Specific Language (Query DSL)
 - Defining custom methods (low-level coding)

www.luv2code.com/spring-data-jpa-defining-custom-queries

Spring Data REST in Spring Boot



Spring Data JPA

- Earlier, we saw the magic of Spring Data JPA
- This helped to eliminate boilerplate code

Before Spring Data JPA

```
public interface EmployeeDAO {  
    public List<Employee> findAll();  
    public Employee findById(int theId);  
    public void save(Employee theEmployee);  
    public void delete(int theId);  
}
```

```
@Repository  
public class EmployeeDAOImpl implements EmployeeDAO {  
    private EntityManager em;  
    @Autowired  
    public EmployeeDAOImpl(EntityManager em) {  
        this.em = em;  
    }  
    @Override  
    public List<Employee> findAll() {  
        // create TypedQuery  
        TypedQuery<Employee> query = em.createQuery("select e from Employee e", Employee.class);  
        // return employee  
        return query.getResultList();  
    }  
    @Override  
    public Employee findById(int theId) {  
        // get employee  
        Employee theEmployee = em.find(Employee.class, theId);  
        // return employee  
        return theEmployee;  
    }  
}
```

2 Files
30+ lines of code

After Spring Data JPA

```
public interface EmployeeRepository extends JpaRepository<Employee, Integer> {  
    // that's it ... no need to write any code LOL!  
}
```

1 File
3 lines of code!

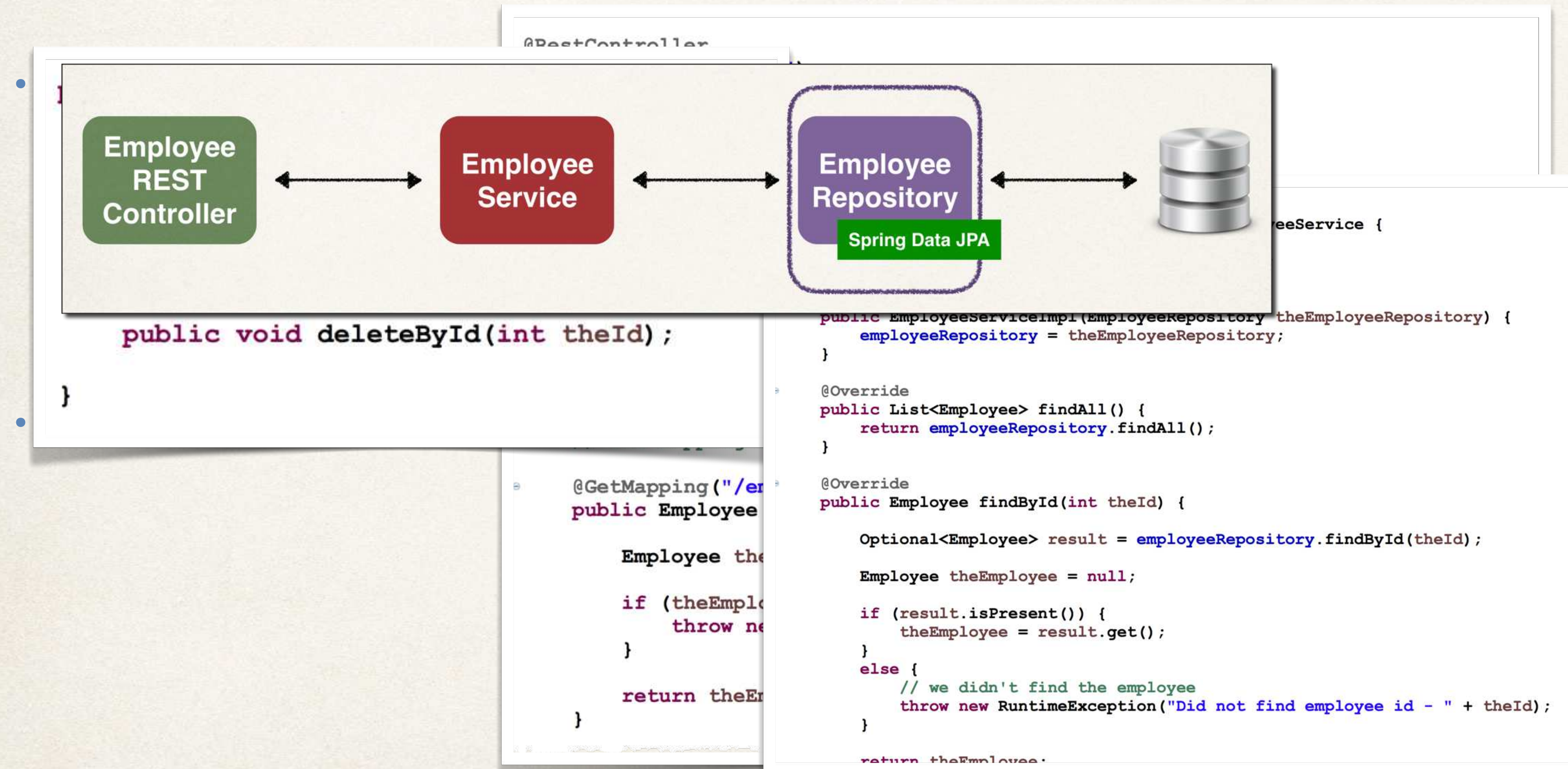
No need for
implementation class



Hmmm ...
Can this apply to REST APIs?

The Problem

- We saw how to create a REST API for **Employee**



My Wish

- I wish we could tell Spring:

Create a REST API for me

Use my existing JpaRepository (entity, primary key)

Give me all of the basic REST API CRUD features for free

Spring Data REST - Solution

- Spring Data REST is the solution!!!!

<https://spring.io/projects/spring-data-rest>

- Leverages your existing **JpaRepository**
- Spring will give you a REST CRUD implementation for FREE like MAGIC!!
- Helps to minimize boiler-plate REST code!!!
- No new coding required!!!



Spring Data REST - How Does It Work?

- Spring Data REST will scan your project for **JpaRepository**
- Expose REST APIs for each entity type for your **JpaRepository**

```
public interface EmployeeRepository extends JpaRepository<Employee, Integer> {  
  
}
```

REST Endpoints

- By default, Spring Data REST will create endpoints based on entity type
- Simple pluralized form
 - First character of Entity type is lowercase
 - Then just adds an "s" to the entity

*More on
plural forms in
later video*

```
public interface EmployeeRepository extends JpaRepository<Employee, Integer> {  
}
```

/employees

Development Process

Step-By-Step

1. Add Spring Data REST to your Maven POM file

That's it!!!

Absolutely NO CODING required

Step 1: Add Spring Data REST to POM file

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-data-rest</artifactId>
</dependency>
```

That's it!!!

Absolutely NO CODING required

**Spring Data REST will
scan for JpaRepository**

HTTP Method		CRUD Action
POST	/employees	<u>C</u> reate a new employee
GET	/employees	<u>R</u> ead a list of employees
GET	/employees/{employeeId}	<u>R</u> ead a single employee
PUT	/employees/{employeeId}	<u>U</u> pdate an existing employee
DELETE	/employees/{employeeId}	<u>D</u> delete an existing employee

Get these REST
endpoints for free



In A Nutshell

For Spring Data REST, you only need 3 items

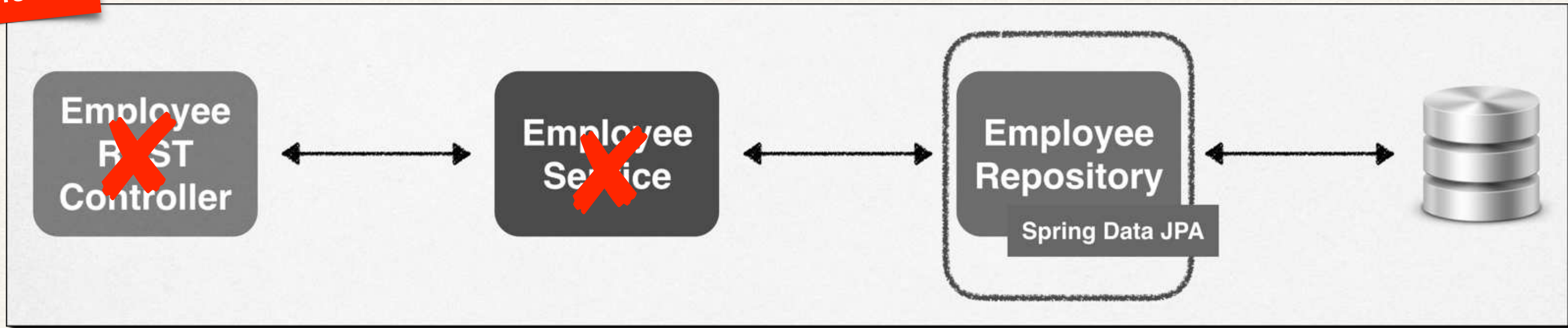
We already have these two

1. Your entity: **Employee**
2. JpaRepository: **EmployeeRepository** extends **JpaRepository**
3. Maven POM dependency for: **spring-boot-starter-data-rest**

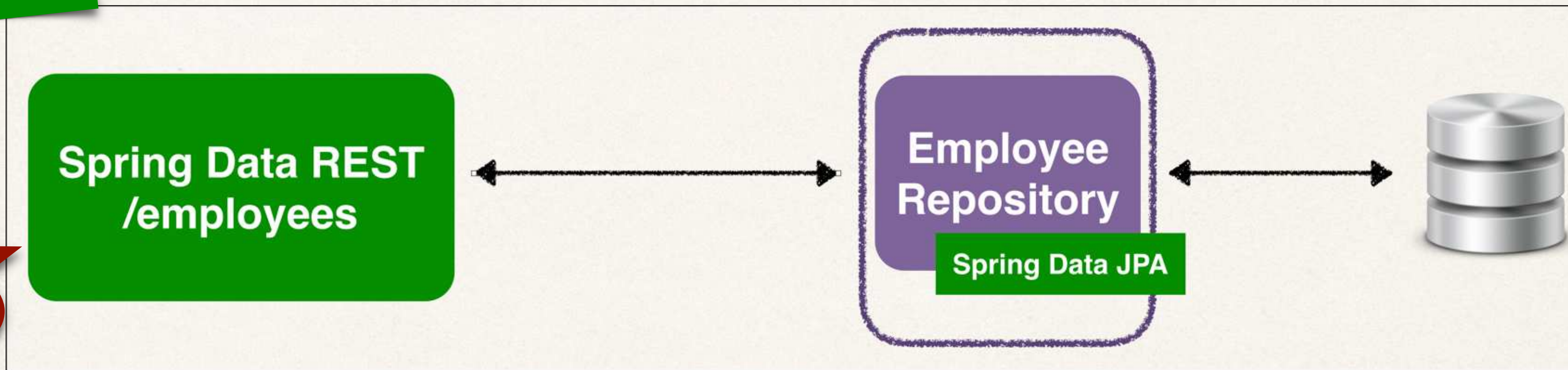
Only item that is new

Application Architecture

Before



After



Get these REST endpoints for free

Minimized Boilerplate Code

Before Spring Data REST

```
@RestController
@RequestMapping("/api")
public class EmployeeRestController {

    private EmployeeService employeeService;

    @Autowired
    public EmployeeRestController(EmployeeService theEmployeeService) {
        employeeService = theEmployeeService;
    }

    // expose "/employees" and return list of
    @GetMapping("/employees")
    public List<Employee> findAll() {
        return employeeService.findAll();
    }

    // add mapping for GET /employees/{id}
    @GetMapping("/employees/{id}")
    public Employee getEmployeeById(int theId) {
        Employee theEmployee = employeeService.findById(theId);

        if (theEmployee == null) {
            throw new RuntimeException("Employee not found");
        }

        return theEmployee;
    }
}
```

```
public interface EmployeeService {

    public List<Employee> findAll();

    public Employee findById(int theId);

    public Employee findById(int theId) implements EmployeeService {

        @Autowired
        private EmployeeRepository employeeRepository;

        @Override
        public List<Employee> findAll() {
            return employeeRepository.findAll();
        }

        @Override
        public Employee findById(int theId) {
            Employee result = employeeRepository.findById(theId);

            if (result == null) {
                throw new RuntimeException("Employee not found");
            }

            return result.get();
        }

        // add the employee
        public void save(Employee theEmployee) {
            employeeRepository.save(theEmployee);
        }

        public void delete(int theId) {
            employeeRepository.deleteById(theId);
        }
    }
}
```

3 files
100+ lines of code

After Spring Data REST

```
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-data-rest</artifactId>
</dependency>
```

That's it!!!

Absolutely NO CODING required

HTTP Method	Endpoint	CRUD Action
POST	/employees	Create a new employee
GET	/employees	Read a list of employees
GET	/employees/{employeeId}	Read a single employee
PUT	/employees/{employeeId}	Update an existing employee
DELETE	/employees/{employeeId}	Delete an existing employee

Spring Data REST will scan for JpaRepository

Get these REST endpoints for free



HATEOAS

- Spring Data REST endpoints are HATEOAS compliant
- **HATEOAS:** Hypermedia as the Engine of Application State
- Hypermedia-driven sites provide information to access REST interfaces
- Think of it as meta-data for REST data

<https://spring.io/projects/spring-hateoas>

HATEOAS

- Spring Data REST response using HATEOAS
- For example REST response from: **GET /employees/3**

Get single
employee

Response

```
{
  "firstName": "Avani",
  "lastName": "Gupta",
  "email": "avani@luv2code.com",
  "_links": {
    "self": {
      "href": "http://localhost:8080/employees/3"
    },
    "employee": {
      "href": "http://localhost:8080/employees/3"
    }
  }
}
```

Employee data

Response meta-data
Links to data

HATEOAS

- For a collection, meta-data includes page size, total elements, pages etc
- For example REST response from: **GET /employees**

Get list of employees

Response

```
{
  "_embedded": {
    "employees": [
      {
        "firstName": "Leslie",
        ...
      },
      ...
    ]
  },
  "page": {
    "size": 20,
    "totalElements": 5,
    "totalPages": 1,
    "number": 0
  }
}
```

JSON Array of employees

Response meta-data
Information about the page

*More on
configuring page
sizes later*

HATEOAS

- For details on HATEOAS, see

<https://spring.io/projects/spring-hateoas>

- HATEOAS uses Hypertext Application Language (HAL) data format
- For details on HAL, see

https://en.wikipedia.org/wiki/Hypertext_Application_Language

Advanced Features

- Spring Data REST advanced features
 - Pagination, sorting and searching
 - Extending and adding custom queries with JPQL
 - Query Domain Specific Language (Query DSL)

<https://spring.io/projects/spring-data-rest>

Spring Data REST Configuration, Pagination and Sorting



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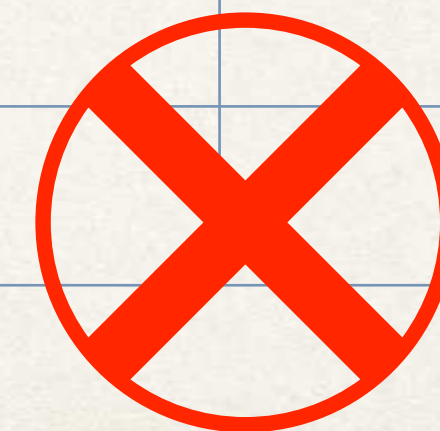


/employees

Pluralized Form

- Spring Data REST pluralized form is VERY simple
 - Just adds an "s" to the entity
- The English language is VERY complex!
 - Spring Data REST does NOT handle

Singular	Plural
Goose	Geese
Person	People
Syllabus	Syllabi
...	...



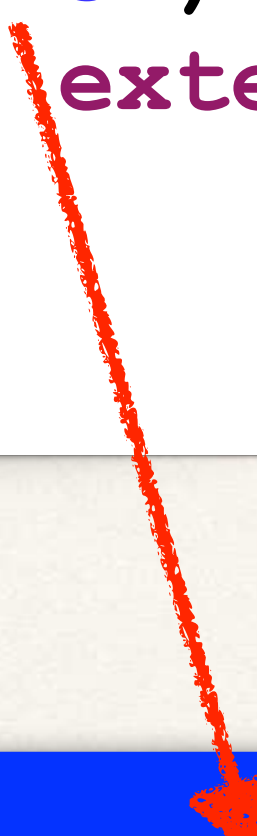
Problem

- Spring Data REST does not handle complex pluralized forms
 - In this case, you need to specify plural name
- What if we want to expose a different resource name?
 - Instead of **/employees** ... use **/members**

Solution

- Specify plural name / path with an annotation

```
@RepositoryRestResource(path="members")  
public interface EmployeeRepository extends JpaRepository<Employee, Integer> {  
}
```



<http://localhost:8080/members>

Pagination

- By default, Spring Data REST will return the first 20 elements
 - Page size = 20
- You can navigate to the different pages of data using query param

```
http://localhost:8080/employees?page=0
```

```
http://localhost:8080/employees?page=1
```

```
...
```

**Pages are
zero-based**

Spring Data REST Configuration

- Following properties available: application.properties

Name	Description
<code>spring.data.rest.base-path</code>	Base path used to expose repository resources
<code>spring.data.rest.default-page-size</code>	Default size of pages
<code>spring.data.rest.max-page-size</code>	Maximum size of pages
...	...

More properties available
www.luv2code.com/spring-boot-props

`spring.data.rest.*`

Sample Configuration

<http://localhost:8080/magic-api/employees>

File: application.properties

```
spring.data.rest.base-path=/magic-api  
spring.data.rest.default-page-size=50
```

Returns 50
elements per page

Sorting

- You can sort by the property names of your entity
 - In our Employee example, we have: **firstName**, **lastName** and **email**
- Sort by last name (ascending is default) <http://localhost:8080/employees?sort=lastName>
- Sort by first name, descending <http://localhost:8080/employees?sort=firstName,desc>
- Sort by last name, then first name, ascending <http://localhost:8080/employees?sort=lastName,firstName,asc>